

UNSW
AUSTRALIA

SCHOOL OF ELECTRICAL ENGINEERING AND TELECOMMUNICATIONS

Resource Allocation for Integrated Sensing and Communication (ISAC) in Cell-Free Massive MIMO Systems

by

Xinpeng Yu

Student ID: z5615351

Thesis submitted as a requirement for the degree
Master of Engineering (Telecommunications)

Submitted: August 2026

Supervisor: Derrick Wing Kwan Ng

Co-supervisor: Shaokang Hu

Abstract

This thesis studies robust power-minimizing resource allocation for a cell-free massive MIMO integrated sensing and communication network. Distributed access points cooperatively serve communication users and transmit target-specific sensing waveforms under norm-bounded channel-state uncertainty. The proposed formulation jointly guarantees worst-case communication SINR, sensing SINR, and a matrix PCRB constraint while selecting a fixed-size sensing cluster for each active target. In contrast to formulations that credit all transmit covariance to sensing, the tracking FIM here is generated only by the known waveform dedicated to the corresponding target. The resulting mixed-integer non-convex problem is transformed through covariance lifting, an S-Procedure LMI, an exact Schur-complement PCRB LMI, and joint rank-one/binary DC penalties. A topology-aware SCA recovery procedure finally re-solves fixed binary candidates and validates the extracted physical beamformers. Monte-Carlo evaluation over common random scenarios shows that the proposed recovery preserves physical feasibility in every tested AP-count, load, area, budget, and stress-geometry configuration, whereas a nearest-AP topology remains feasible in only about one quarter of crowded-target scenarios.

Acknowledgements

This thesis marks the culmination of my master's journey—a destination I could never have reached alone.

First, I thank my co-supervisor, Doctor Shaokang Hu, for his meticulous guidance throughout the past two years. From research ideas to experimental details, his patience and expertise shaped every stage of this work.

I am equally grateful to my supervisor, Associate Professor Derrick Wing Kwan Ng, for his invaluable guidance in shaping the research topic and his unwavering support throughout this journey. His vision and insights kept this research on course.

I thank my family for their unconditional understanding and quiet support. Their encouragement traveled across oceans, becoming my anchor in moments of doubt. This accomplishment belongs to them as much as it does to me.

I am deeply thankful to Carol Ma, whose companionship made these two years in Sydney not only productive but truly joyful. In times of pressure, you made everything feel lighter.

Finally, I thank my fellow students in Sydney—Tingxu Huang, Chongyi Li, Qianyu Xu, and others—for their academic camaraderie and everyday support. Their presence turned a foreign city into a place of belonging.

And to Sydney—to its vast, unhurried sea, which reminded me, simply by being there, that some things need only be witnessed, not solved.

To have met you all—and this city—along this path is my greatest fortune.

Declaration of AI Use

The author acknowledges the use of Kimi, ChatGPT, and Gemini for language polishing of select passages and code assistance in specific sections. All core research, critical code, data analysis, and conclusions were completed independently by the author. All AI-assisted content has been reviewed and verified by the author, who assumes full responsibility.

Abbreviations

6G Sixth-Generation wireless networks

AP Access Point

CPU Central Processing Unit

CRB Cramér–Rao Bound

CSI Channel State Information

DC Difference of Convex

DCP Difference-of-Convex Program

FIM Fisher Information Matrix

ISAC Integrated Sensing and Communication

KKT Karush–Kuhn–Tucker

LMI Linear Matrix Inequality

MIMO Multiple-Input Multiple-Output

MINLP Mixed-Integer Non-Linear Programming

MISO Multiple-Input Single-Output

mmWave Millimeter Wave

OTFS Orthogonal Time Frequency Space

PCRB Posterior Cramér–Rao Bound

PL Path Loss

PoD Probability of Detection

PSD Positive Semidefinite

QoS Quality of Service

SCA Successive Convex Approximation

SDP Semidefinite Programming

SDR Semidefinite Relaxation

SINR Signal-to-Interference-plus-Noise Ratio

STAR-RIS Simultaneously Transmitting and Reflecting Reconfigurable Intelligent Surface

Notation

Bold lower-case and upper-case letters denote vectors and matrices, respectively. $(\cdot)^T$, $(\cdot)^H$, $\mathrm{Tr}(\cdot)$, and $\|\cdot\|_2$ denote transpose, Hermitian transpose, trace, and the Euclidean norm.

$\mathcal{M}, \mathcal{K}, \mathcal{P}$ Sets of APs, communication users, and sensing targets.

M, K, P Numbers of APs, communication users, and sensing targets.

N_t, $N = MN_t$ Antennas per AP and total stacked transmit dimension.

$\mathcal{P}^{\mathrm{active}}$ Targets requiring sensing service in the current scheduling interval.

N_{req} Number of APs authorized for each active target.

$\boldsymbol{w}_{m,k}$, $\boldsymbol{w}_k$ Beamformer from AP m to user k, and its network-wide stacked form.

$\boldsymbol{W}_k$ Lifted communication covariance, $\boldsymbol{w}_k \boldsymbol{w}_k^H$.

$\boldsymbol{S}_p$ Network-wide covariance of the dedicated sensing waveform for target p.

$\boldsymbol{R}_X$ Total transmit covariance, $\sum_k \boldsymbol{W}_k + \sum_p \boldsymbol{S}_p$.

b_{mp} Binary authorization indicator for AP m to allocate dedicated sensing power to target p.

$\boldsymbol{E}_m$ Diagonal matrix selecting the antenna block of AP m.

$\hat{\boldsymbol{h}}_k$, $\Delta \boldsymbol{h}_k$, $\boldsymbol{h}_k$ Nominal channel, channel error, and uncertain stacked channel of user k.

ϵ_h Relative radius of the norm-bounded CSI uncertainty set.

$\boldsymbol{g}_p$, $\boldsymbol{D}_p$ Sensing response vector and its derivative matrix for target p.

$\boldsymbol{J}_p^{\mathrm{data}}$, $\boldsymbol{M}_p$ Data FIM and auxiliary matrix used in the PCRB Schur-complement LMI.

γ_k, γ_S^{PoD} Required communication SINR and sensing-SINR threshold.

$\Gamma_{\mathrm{Track},p}$ Maximum permitted trace PCRB for target p.

σ_c^2, σ_s^2 Communication-noise and sensing-noise powers.

$P_{\max}$ Per-AP total transmit-power budget.

μ_k Nonnegative S-Procedure multiplier for user k.

$\boldsymbol{\theta}_p$, N_θ Target-state vector of target p and its dimension.

$\boldsymbol{S}_\Sigma$ Aggregate dedicated sensing covariance, $\sum_p \boldsymbol{S}_p$.

$\boldsymbol{C}_p$ CRB matrix $(\boldsymbol{J}_p^{\text{data}})^{-1}$.

$\hat{\boldsymbol{H}}_k(\Delta\boldsymbol{h}_k)$ Rank-one uncertain channel matrix $(\hat{\boldsymbol{h}}_k + \Delta\boldsymbol{h}_k)(\hat{\boldsymbol{h}}_k + \Delta\boldsymbol{h}_k)^H$.

$\boldsymbol{A}_k$ SINR combination matrix $\frac{1}{\gamma_k}\boldsymbol{W}_k - \sum_{j\neq k}\boldsymbol{W}_j - \sum_p \boldsymbol{S}_p$.

$\lambda_{\max}(\cdot)$, $\boldsymbol{u}_{\max,k}$ Largest eigenvalue and its eigenvector for $\boldsymbol{W}_k$.

$\rho(\boldsymbol{W}_k)$ Rank-deficiency measure $\text{Tr}(\boldsymbol{W}_k) - \lambda_{\max}(\boldsymbol{W}_k)$.

$g(b)$ Binarity-deficiency measure $b - b^2$.

η_{rank}, η_b Penalty factors for the rank-one and binarity DC terms.

ε, $T_{\max}$ SCA stopping tolerance and maximum iteration count.

L Number of Gaussian randomization candidates.

Contents

Acknowledgements **1**

Declaration of AI Use **2**

Abbreviations **3**

Notation **5**

1 Introduction **9**
1.1 Motivation 9
1.2 Research Gap, Problem, and Contributions 9
1.3 Thesis Organization 10

2 Literature Review **11**
2.1 From Coexistence to Cooperative ISAC 11
2.2 Design Requirements for Robust Multi-Target Sensing 12
2.2.1 Communication QoS under Imperfect CSI 12
2.2.2 Target-Specific Detection and Tracking Accuracy . . . 12
2.2.3 Sensing Topology and Physical Recovery 12
2.3 Research Gap and Thesis Position 13

3 System Model **14**
3.1 Network and Signalling Model 14
3.2 Robust Communication Model 16
3.3 Target-Specific Sensing Model 16

4 Optimization Formulation and Solution **19**
4.1 Problem Formulation 19
4.2 Solution of the Optimization Problem 23
A. Problem Transformation (P2) 23
B. Semi-Definite Relaxation 24
C. S-Procedure 25

D. PCRB Constraint: Schur Complement LMI Transformation 26
E. AP–Target Soft Assignment: Continuous Relaxation with DC Penalty . . . 27
F. Consolidation (P3): Final Form with Dual DC Penalties . . 28
4.3 Joint DC Penalty SCA Framework: Rank-One and Binary Recovery . . . 29
A. Unified Characterization of the Dual DC Penalties 29
B. SCA Linearization 30
C. Baseline: Gaussian Randomization / Eigen-Extraction . . . 31
D. Candidate Re-Solution and Physical Validation 33
4.4 Computational Complexity and Scalability 35
4.5 Link to Physical Evaluation 36

5 Numerical Evaluation and Results Analysis 37
5.1 Evaluation Protocol, Metrics, and Baselines 37
5.2 Topology-Recovery Gains in Ordinary and Stress Geometries . 38
5.3 Robustness–Power Trade-Off under CSI Uncertainty 41
5.4 Recovery Mechanism and DC-SCA Stabilization 42
5.5 Larger-Scale Validation and Computational Cost 43
5.6 Factor Sweeps, Paired Statistics, and QoS Margins 44
5.7 Summary of Findings 48

6 Conclusion and Future Work 49
6.1 Conclusion 49
6.2 Future Work 50

Chapter 1

Introduction

1.1 Motivation

Sixth-generation wireless networks must provide reliable connectivity and timely awareness of the surrounding environment for applications such as industrial automation, intelligent transport, and autonomous aerial systems [1, 2]. [3, 4] Integrated sensing and communication (ISAC) addresses this requirement by sharing spectrum, radio hardware, and spatial degrees of freedom between data delivery and sensing. The resulting benefit is compelling, but the shared resources also create a fundamental allocation problem: power assigned to communication, dedicated sensing, and interference protection cannot be chosen independently.

Cell-free massive MIMO makes this problem both more promising and more challenging. A central processing unit can coordinate distributed access points (APs) to serve users cooperatively, thereby creating spatial diversity for communication and sensing without fixed cell boundaries [5, 6]. [7, 8] However, each AP has a finite transmit-power budget. Selecting which APs may illuminate each target therefore changes the sensing geometry and the remaining power available for globally cooperative communication. The decision must also remain reliable when the communication channel state information (CSI) is imperfect.

1.2 Research Gap, Problem, and Contributions

Existing cooperative ISAC formulations establish the value of spatial cooperation, yet three connected gaps remain for the design considered here.

First, a sensing association is often difficult to interpret when the model does not distinguish a target's dedicated sensing waveform from the globally cooperative communication signal. Second, perfect-CSI formulations can lose their communication quality-of-service (QoS) guarantee under channel error. Third, a relaxed semidefinite or continuous association solution is not itself a physical beamforming and topology design; rank-one and binary recovery must be assessed after the relaxation is solved.

This thesis addresses these gaps through robust downlink resource allocation for a cell-free ISAC network. The objective is to minimize total transmit power while satisfying worst-case communication SINR, target detection, and tracking accuracy requirements. The central design choice is a known, target-specific sensing covariance for each target. Its binary AP–target association variable authorizes dedicated sensing power only; it does not gate the globally cooperative communication beamformers. This separation makes the topology decision physically interpretable and permits each target's Fisher information and PCRB requirement to be evaluated from its own probing covariance.

1.3 Thesis Organization

Chapter 2 narrows the literature from cooperative ISAC to the four design requirements that motivate this thesis: cooperative resource allocation, robust QoS, multi-target tracking accuracy, and recoverable sensing topology. Chapter 3 then defines the network roles, the robust communication model, and the target-specific sensing model. Chapter 4 formulates the power-minimization problem and derives the lifted LMI and dual-DC recovery framework. Chapter 5 evaluates the proposed workflow through topology, robustness, recovery, and scalability evidence. Finally, Chapter 6 answers the research problem from the accumulated evidence and identifies the validation work that remains.

Chapter 2

Literature Review

This chapter positions the thesis around the decisions that must be made by a robust cell-free ISAC transmitter. Rather than treating ISAC architectures, sensing metrics, and optimization methods as separate topics, the review follows the same chain as the proposed design: cooperation creates useful spatial degrees of freedom, QoS protection constrains their use, target-state accuracy determines the sensing requirement, and topology recovery determines whether the computed solution is physically implementable.

2.1 From Coexistence to Cooperative ISAC

ISAC emerged from the observation that wireless communication and sensing rely on the same electromagnetic propagation, spectrum, and radio hardware. Early single-base-station systems therefore studied waveform sharing, communication– sensing trade-offs, and self-interference management [9, 10, 11]. [3, 4] These studies establish the core trade-off, but a single transmitter provides limited geometric diversity and can be vulnerable to cell-edge blockage.

Cooperative and multistatic ISAC extend the sensing and communication aperture across multiple base stations or APs. Spatially distributed observations can improve coverage, localization geometry, and interference management, provided that the transmitters are coordinated [12, 13, 14]. [7, 8, 15] Cell-free massive MIMO takes this idea further: a large set of distributed APs is centrally coordinated without fixed cell boundaries [5, 6]. [7, 16] The same cooperation that improves service flexibility also couples every AP's communication and sensing power decisions. This is the architectural setting of the present thesis.

2.2 Design Requirements for Robust Multi-Target Sensing

2.2.1 Communication QoS under Imperfect CSI

Communication QoS is commonly expressed through rate or SINR constraints. Many ISAC formulations assume perfect CSI to simplify joint beamforming, but estimation error, feedback delay, and quantization make this assumption fragile in practice [17, 18]. [19, 20] For a cell-free transmitter, channel uncertainty is particularly consequential: the dedicated sensing covariances and multi-user communication beams share the same AP power budgets and jointly determine the user interference level. This motivates a norm-bounded worst-case SINR requirement rather than a nominal-channel guarantee, in line with worst-case robust multiuser designs developed for related downlink settings [21].

2.2.2 Target-Specific Detection and Tracking Accuracy

Sensing designs are often evaluated with sensing SINR and Cramér–Rao-bound (CRB) metrics. Sensing SINR is a detection-related proxy, whereas the CRB measures a lower bound on estimation error and is consequently useful for localization and tracking-oriented design [14, 22]. [23] For multiple targets, however, the sensing resource assigned to one target should not be credited to another merely because the network uses a shared transmit infrastructure. The present model therefore assigns a known sensing covariance to each target and forms its FIM and matrix PCRB from that covariance alone. This choice makes the association and tracking requirements refer to the same physical sensing resource.

2.2.3 Sensing Topology and Physical Recovery

Cell-free systems can authorize subsets of APs to allocate a sensing waveform for a target, trading sensing geometry against finite AP power and implementation cost. This turns sensing-topology authorization into a mixed-integer design variable rather than a post-processing detail. Semidefinite relaxation is valuable for handling beamforming structure, but a relaxed covariance or a continuous association weight does not by itself establish an implementable design. A complete workflow must recover rank-one beamformers and binary associations, then test the recovered variables against the original communication, sensing, tracking, and power constraints.

Recent work also broadens ISAC across waveform, hardware, and surface-assisted architectures: OTFS exploits delay–Doppler structure for high-mobility sensing and communication [24]; one-bit transceiver design addresses low-resolution mmWave hardware [25]; and STAR-RIS and holographic surfaces provide relevant near-field extensions [26, 27]. Reconfigurable-antenna ISAC variants are also emerging, including fluid-antenna nearshore systems with joint beamforming and port selection [28], pinching-antenna ISAC with waveguide mode selection [29], and robust blockage-aware pinching-antenna communication [30]. On the communication side, pinching-antenna architectures have been studied for multi-waveguide NOMA and broadcast resource allocation [31, 32], and movable-antenna arrays for energy-efficient and RSMA-based networks [33, 34]. Cooperation has further been extended to relay-assisted nearshore ISAC [35], and security-oriented designs jointly protect communication and sensing in MC-NOMA ISAC systems [36]. At the system layer, sensing-aware status updating offers a complementary scheduling view [37], and power-efficient allocation with coupled discrete and continuous decisions has been demonstrated in IRS-assisted UAV-NOMA networks [38]. These directions motivate the scope of this thesis but are not assumed by its conventional multi-antenna AP signal model.

2.3 Research Gap and Thesis Position

The preceding literature motivates a precise gap at the intersection of the four requirements. Cooperative ISAC provides the relevant spatial degrees of freedom; robust QoS prevents their allocation from relying on nominal CSI; target-specific FIM/PCRB modelling makes dedicated sensing participation meaningful; and recovery-aware optimization separates a feasible relaxed point from a validated physical design. This thesis combines these elements in one power-minimizing cell-free ISAC formulation. Chapter 3 defines the required resource separation, and Chapter 4 develops the corresponding robust and recovery-aware optimization method.

Chapter 3

System Model

The model is organized around a deliberate resource separation. Communication is a network-wide cooperative service, whereas sensing is accounted for through a distinct, known covariance for each target. The AP–target association will therefore answer a narrow physical question—which APs may allocate dedicated sensing power to a target—without artificially restricting the communication beamforming support. The following sections introduce this separation in the same order in which it enters the optimization problem: network resources, robust communication protection, and target-specific sensing accuracy.

3.1 Network and Signalling Model

We consider a downlink cell-free integrated sensing and communication (ISAC) network in which a central processing unit coordinates M distributed access points (APs). Each AP has N_t transmit antennas and jointly serves K single-antenna communication users while illuminating P sensing targets. The AP, user, and target index sets are denoted by $\mathcal{M} = \{1, \ldots, M\}$, $\mathcal{K} = \{1, \ldots, K\}$, and $\mathcal{P} = \{1, \ldots, P\}$, respectively. A subset $\mathcal{P}^{\text{active}} \subseteq \mathcal{P}$ contains the targets that require sensing service in the current scheduling interval. All experiments in this thesis schedule every target, i.e. $\mathcal{P}^{\text{active}} = \mathcal{P}$'; the subset notation is retained so that the model extends to partial activation without modification. The model is instantaneous within one coherence block; target-state prediction across blocks is outside the scope of the resource-allocation problem.

Let $\boldsymbol{w}_{m,k} \in \mathbb{C}^{N_t}$ be the communication beamformer from AP m to user k. To make sensing participation explicit, a distinct zero-mean sensing waveform is assigned to every target. Its network-wide stacked covariance is $\boldsymbol{S}_p \succeq \mathbf{0}$,

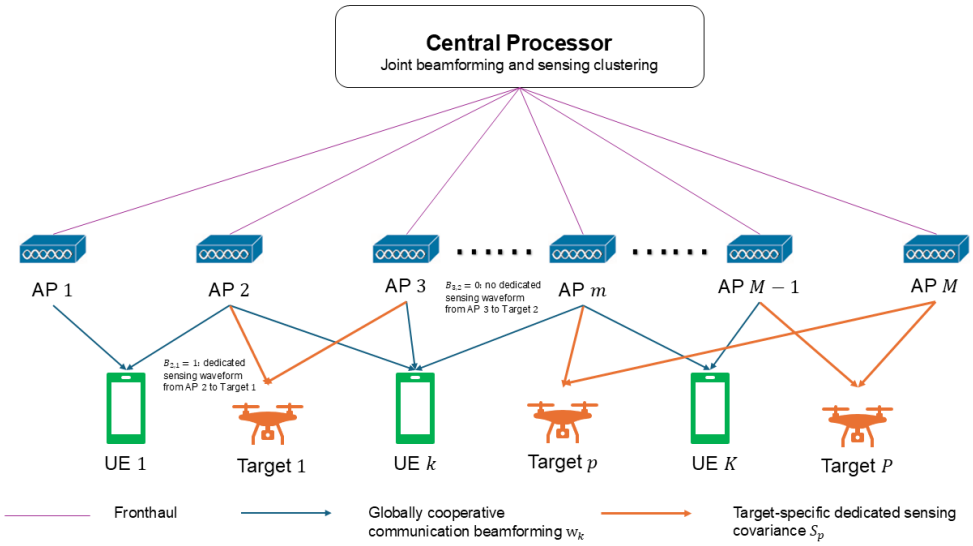


Figure 3.1: Cell-free ISAC architecture. The CPU coordinates distributed APs through fronthaul/control links (magenta). APs jointly serve users through globally cooperative communication beamforming (blue), whereas only associations with $b_{mp} = 1$ authorize the target-specific dedicated sensing covariance $\boldsymbol{S}_p$ (orange). The generic-index illustration illustrates an inactive association, $b_{3,2} = 0$, which denies AP 3 dedicated sensing power for target 2 only; it does not prevent AP 3 from contributing to communication.

where the stacked transmit dimension is $N = MN_t$. Define the stacked communication beamformer

$$\boldsymbol{w}_k \triangleq [\boldsymbol{w}_{1,k}^{\mathrm{T}}, \ldots, \boldsymbol{w}_{M,k}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^N, \tag{3.1}$$

and its covariance $\boldsymbol{W}_k = \boldsymbol{w}_k \boldsymbol{w}_k^{\mathrm{H}}$. The total transmit covariance is therefore

$$\boldsymbol{R}_X = \sum_{k \in \mathcal{K}} \boldsymbol{W}_k + \sum_{p \in \mathcal{P}} \boldsymbol{S}_p. \tag{3.2}$$

Let $\boldsymbol{E}_m \in \mathbb{R}^{N \times N}$ select the N_t-antenna block of AP m. The per-AP hardware constraint is

$$\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X) \leq P_{\max}, \qquad \forall m \in \mathcal{M}. \tag{3.3}$$

3.2 Robust Communication Model

The nominal stacked downlink channel of user k is $\hat{\boldsymbol{h}}_k \in \mathbb{C}^N$. Imperfect channel state information (CSI) is represented by the norm-bounded uncertainty model

$$\boldsymbol{h}_k = \hat{\boldsymbol{h}}_k + \Delta \boldsymbol{h}_k, \qquad \|\Delta \boldsymbol{h}_k\|_2 \leq \epsilon_h \|\hat{\boldsymbol{h}}_k\|_2. \tag{3.4}$$

Let $\boldsymbol{S}_\Sigma \triangleq \sum_{p \in \mathcal{P}} \boldsymbol{S}_p$ denote the aggregate dedicated sensing covariance. It is treated as interference at a communication user. This conservative and implementable assumption avoids requiring receiver-side cancellation of the sensing waveforms. It also provides a worst-case performance bound: any receiver-side exploitation of the known sensing waveforms can only improve the achieved communication SINR, so a design certified under this model remains valid whenever such cancellation becomes available. With $\boldsymbol{h}_k$ from (3.4), the worst-case downlink SINR requirement is

$$\min_{\|\Delta \boldsymbol{h}_k\|_2 \leq \epsilon_h \|\hat{\boldsymbol{h}}_k\|_2} \frac{|\boldsymbol{h}_k^{\mathrm{H}} \boldsymbol{w}_k|^2}{\displaystyle\sum_{j \neq k} |\boldsymbol{h}_k^{\mathrm{H}} \boldsymbol{w}_j|^2 + \boldsymbol{h}_k^{\mathrm{H}} \boldsymbol{S}_\Sigma \boldsymbol{h}_k + \sigma_c^2} \geq \gamma_k, \qquad \forall k \in \mathcal{K}, \tag{3.5}$$

where σ_c^2 is the communication-noise power and γ_k is the required SINR of user k.

3.3 Target-Specific Sensing Model

The binary variable $b_{mp} \in \{0, 1\}$ indicates whether AP m is admitted to the sensing cluster of target p. It authorizes dedicated sensing power only; it

does *not* restrict the globally cooperative communication beamformers. For each active target, exactly N_{req} APs are admitted:

$$\sum_{m\in\mathcal{M}} b_{mp} = N_{\mathrm{req}}, \qquad \forall p \in \mathcal{P}^{\mathrm{active}}. \tag{3.6}$$

The target-specific association gate is

$$\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{S}_p) \le P_{\max} b_{mp}, \qquad \forall m \in \mathcal{M},\ p \in \mathcal{P}. \tag{3.7}$$

Consequently, an unselected AP cannot allocate sensing power to target p, whereas a selected AP is not forced to consume a minimum amount of power. In particular, (3.7) is not a gate on communication power and must be read jointly with the total hardware limit (3.3).

Let $\boldsymbol{g}_p \in \mathbb{C}^N$ be the effective sensing steering channel of target p. The target-specific sensing SINR is

$$\mathrm{SINR}_{S,p} = \frac{\boldsymbol{g}_p^{\mathrm{H}} \boldsymbol{S}_p \boldsymbol{g}_p}{\sigma_s^2} \ge \gamma_S^{\mathrm{PoD}}, \qquad \forall p \in \mathcal{P}^{\mathrm{active}}, \tag{3.8}$$

where σ_s^2 is the sensing-noise power and γ_S^{PoD} is the detection-related threshold.

For a target state $\boldsymbol{\theta}_p \in \mathbb{R}^{N_\theta}$, let $\boldsymbol{D}_p = \partial \boldsymbol{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{N\times N_\theta}$ be the effective response derivative. Under the equivalent MISO sensing observation model, the data Fisher information matrix (FIM) is

$$\boldsymbol{J}_p^{\mathrm{data}}(\boldsymbol{S}_p) = \frac{2}{\sigma_s^2} \mathrm{Re}\big\{ \boldsymbol{D}_p^{\mathrm{H}} \boldsymbol{S}_p \boldsymbol{D}_p \big\}. \tag{3.9}$$

Only the known waveform covariance $\boldsymbol{S}_p$ receives sensing-information credit. Communication covariances and waveforms dedicated to other targets are not counted in (3.9). Tracking accuracy is constrained by the matrix PCRB/CRB requirement below. Within the single-interval model of Section 3.1, *tracking accuracy* denotes the instantaneous state-estimation precision certified within one coherence block; no temporal recursion is implied, and target-state prediction across blocks is deferred to Chapter 6. The requirement is

$$\mathrm{Tr}\big((\boldsymbol{J}_p^{\mathrm{data}})^{-1}\big) \le \Gamma_{\mathrm{Track},p}, \qquad \forall p \in \mathcal{P}^{\mathrm{active}}. \tag{3.10}$$

For $N_\theta = 1$, this reduces to the scalar lower bound $J_p^{\mathrm{data}} \ge 1/\Gamma_{\mathrm{Track},p}$; for multi-dimensional states, the full matrix form is retained. The equivalent semidefinite form used to solve (3.10) is developed in Chapter 4.

Design implication. The model couples the three services through the common AP-power budget but keeps their roles distinguishable: communication is protected for every admissible CSI error, while a target receives sensing credit only from its own authorized covariance. Chapter 4 turns these requirements into the power-minimization problem and identifies the robust, matrix, rank, and binary structures that prevent a direct convex solution.

Chapter 4

Optimization Formulation and Solution

Chapter 3 identifies the physical allocation question: minimize total transmit power while preserving globally cooperative communication, target-specific sensing quality, and AP-level implementability. This chapter keeps that interpretation visible throughout the transformation. Problem (P1) states the original binary physical design; (P2) lifts the beamforming variables and relaxes only the binary box for optimization; and (P3) collects the robust and tracking LMIs with the DC penalties used for candidate recovery. Thus, an LMI-feasible point is an intermediate optimization object, while a recovered design is accepted only after the physical checks used in Chapter 5.

4.1 Problem Formulation

Target-Centric Dedicated Sensing Associations. The binary variable $b_{mp} \in \{0, 1\}$ is a sensing-cluster admission indicator: it specifies whether AP m is authorized to allocate dedicated sensing resources to target p. Let $\boldsymbol{S}_p \succeq \mathbf{0}$ denote the covariance of the corresponding waveform. When $b_{mp} = 1$, AP m may allocate transmit power to $\boldsymbol{S}_p$; when $b_{mp} = 0$, its corresponding covariance block is forced to zero. Thus, $b_{mp} = 1$ is an authorization rather than a minimum-power guarantee, and realized participation is quantified from the nonzero sensing-power blocks after optimization. The indicator does not restrict communication beamforming, which remains globally cooperative. The per-AP transmit-hardware constraint is

$$\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X) \leq P_{\max}, \qquad \forall m \in \mathcal{M}. \tag{4.1}$$

The association constraint and the dedicated sensing resource constraint are

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \qquad \mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{S}_p) \leq P_{\max} b_{mp}, \qquad \forall m, p. \tag{4.2}$$

The cardinality constraint fixes the sensing-cluster size of each target at N_{req}. This equality is a deliberate abstraction of the fronthaul and processing budget of a cell-free deployment: authorizing exactly N_{req} APs per target lets the CPU deterministically reserve the corresponding signalling and per-target processing resources, while the one-sided gate still permits an authorized AP to contribute zero sensing power. A relaxed variant $\sum_m b_{mp} \leq N_{\mathrm{req}}$ would trade this resource determinism for additional topology flexibility and would be meaningful only together with an explicit fronthaul or processing cost.

The total transmit covariance is $\boldsymbol{R}_X = \sum_{k=1}^{K} \boldsymbol{W}_k + \sum_{p=1}^{P} \boldsymbol{S}_p$. To retain a known and target-specific probing signal at the sensing receiver, the PCRB of target p is evaluated only from $\boldsymbol{S}_p$:

$$\boldsymbol{J}_p^{\mathrm{data}}(\boldsymbol{S}_p) = \frac{2}{\sigma_s^2} \mathrm{Re}\big\{ \boldsymbol{D}_p^H \boldsymbol{S}_p \boldsymbol{D}_p \big\} . \tag{4.3}$$

Thus, the association variable controls the dedicated sensing resources that can contribute to the PCRB, whereas communication covariances do not receive PCRB credit. The constraints above are affine in the continuous covariance variables for fixed b_{mp} and linear in b_{mp} for fixed covariances. Unless receiver-side cancellation is explicitly assumed, the aggregate dedicated sensing covariance $\sum_p \boldsymbol{S}_p$ is included as interference in the communication SINR. The Sum Big-M power-gating formulation and derivations based on an FIM that is affine solely in $\boldsymbol{R}_X$ correspond to a different model and should be replaced by the present formulation.

The decision variables of the original problem (P1) are the per-AP beamforming vectors $\boldsymbol{w}_{m,k} \in \mathbb{C}^{N_t}$, the target-specific sensing covariance matrices

$\boldsymbol{S}_p \in \mathbb{H}_+^{MN_t}$, and the binary AP–target association indicators $b_{mp} \in \{0,1\}$:

$$\min_{\substack{\{\boldsymbol{w}_{m,k}\}_{m\in\mathcal{M},k\in\mathcal{K}},\{\boldsymbol{S}_p\}_{p\in\mathcal{P}},\\ \{b_{mp}\}_{m\in\mathcal{M},p\in\mathcal{P}}}} \quad \sum_{m=1}^{M}\sum_{k=1}^{K}\|\boldsymbol{w}_{m,k}\|^2 + \sum_{p=1}^{P}\mathrm{Tr}(\boldsymbol{S}_p) \tag{P1-objective}$$

$$\text{s.t.} \quad \min_{\|\Delta\boldsymbol{h}_k\|\le\epsilon_h\|\hat{\boldsymbol{h}}_k\|} \frac{|\boldsymbol{h}_k^H\boldsymbol{w}_k|^2}{\sum_{j\neq k}|\boldsymbol{h}_k^H\boldsymbol{w}_j|^2 + \boldsymbol{h}_k^H\big(\sum_p \boldsymbol{S}_p\big)\boldsymbol{h}_k + \sigma_c^2} \ge \gamma_k, \quad \forall k, \tag{P1-C1}$$

$$\mathrm{SINR}_{S,p} := \frac{\boldsymbol{g}_p^H\boldsymbol{S}_p\boldsymbol{g}_p}{\sigma_s^2} \ge \gamma_S^{\mathrm{PoD}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, \tag{P1-C2}$$

$$\mathrm{Tr}\left((\boldsymbol{J}_p^{\mathrm{data}}(\boldsymbol{S}_p))^{-1}\right) \le \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, \tag{P1-C3}$$

$$\mathrm{Tr}(\boldsymbol{E}_m\boldsymbol{S}_p) \le P_{\max}\, b_{mp}, \quad \forall m \in \mathcal{M},\, p \in \mathcal{P}, \tag{P1-C4'a}$$

$$\mathrm{Tr}(\boldsymbol{E}_m\boldsymbol{R}_X) \le P_{\max}, \quad \forall m \in \mathcal{M}, \tag{P1-C4'b}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, \tag{P1-C5'}$$

$$\boldsymbol{S}_p \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}, \tag{P1-C6}$$

$$b_{mp} \in \{0,1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \tag{P1-C7'}$$

Here $\boldsymbol{h}_k = \hat{\boldsymbol{h}}_k + \Delta\boldsymbol{h}_k$ is the uncertain stacked channel from (3.4). The cooperative stacked beamforming vector is $\boldsymbol{w}_k \triangleq [\boldsymbol{w}_{1,k}^T, \boldsymbol{w}_{2,k}^T, \ldots, \boldsymbol{w}_{M,k}^T]^T \in \mathbb{C}^{MN_t}$, obtained by stacking the per-AP variables. Further, $\boldsymbol{R}_X \triangleq \sum_{k=1}^{K}\boldsymbol{w}_k\boldsymbol{w}_k^H + \sum_{p=1}^{P}\boldsymbol{S}_p$ (precisely defined in the lifted form of §4.2-A), and $\boldsymbol{E}_m \in \mathbb{R}^{MN_t\times MN_t}$ is the diagonal selection matrix that extracts the antenna block of AP m from $\boldsymbol{R}_X$. The FIM $\boldsymbol{J}_p^{\mathrm{data}}$ in (P1-C3) takes the following original form for a multi-dimensional target state $\boldsymbol{\theta}_p \in \mathbb{R}^{N_\theta}$ (typical ISAC scenarios involve 2D/3D position, or angle+range with $N_\theta \in \{2,3,4\}$):

$$\boldsymbol{J}_p^{\mathrm{data}}(\boldsymbol{S}_p) = \frac{2}{\sigma_s^2}\,\mathrm{Re}\big\{\boldsymbol{D}_p^H\boldsymbol{S}_p\boldsymbol{D}_p\big\} \in \mathbb{H}_+^{N_\theta\times N_\theta}, \tag{4.4}$$

where $\boldsymbol{D}_p \triangleq \partial\boldsymbol{g}_p/\partial\boldsymbol{\theta}_p \in \mathbb{C}^{MN_t\times N_\theta}$ is the effective target-response derivative matrix. The equivalent MISO observation model uses the dedicated sensing covariance directly; hence (4.4) restates (4.3) and is affine in $\boldsymbol{S}_p$ only—the communication covariances $\{\boldsymbol{W}_k\}$ receive no PCRB credit. The CRB matrix is $\boldsymbol{C}_p = (\boldsymbol{J}_p^{\mathrm{data}})^{-1}$ and (P1-C3) requires $\mathrm{Tr}(\boldsymbol{C}_p) \le \Gamma_{\mathrm{Track},p}$. This matrix trace-of-inverse constraint must be retained for $N_\theta \ge 2$; for $N_\theta = 1$, it reduces to

the scalar condition $\boldsymbol{J}_p^{\text{data}} \geq 1/\Gamma$. The association gate (P1-C4'a) and the hardware ceiling (P1-C4'b) are both linear: (P1-C4'a) authorizes AP m to allocate dedicated sensing power to target p up to $P_{\max}$ when $b_{mp} = 1$ and forces the corresponding block of $\boldsymbol{S}_p$ to zero when $b_{mp} = 0$, without gating communication transmission, while (P1-C4'b) caps the total per-AP transmit covariance $\boldsymbol{R}_X$.

Problem (P1) is non-convex. Its non-convexity stems from four independent sources distributed across different constraints. The communication worst-case SINR constraint (P1-C1) carries a fractional quadratic structure: the numerator $|\boldsymbol{h}_k^H \boldsymbol{w}_k|^2$ and the denominator $\sum_{j\neq k} |\boldsymbol{h}_k^H \boldsymbol{w}_j|^2 + \boldsymbol{h}_k^H \big(\sum_p \boldsymbol{S}_p\big)\boldsymbol{h}_k + \sigma_c^2$ are both quadratic functions of the beamforming vectors and the sensing covariances, so their ratio is a non-convex function on the manifold where the denominator is positive [39]. Compounding this, the worst-case quantifier $\forall \|\Delta \boldsymbol{h}_k\| \leq \epsilon_h \|\hat{\boldsymbol{h}}_k\|$ is semi-infinite, and the resulting set $\{\boldsymbol{w} : \min_{\Delta \boldsymbol{h} \in \mathcal{B}_\epsilon} \text{SINR}_k(\Delta \boldsymbol{h}) \geq \gamma_k\}$ is not closed under convex combinations. The position PCRB constraint (P1-C3) requires $\text{Tr}((\boldsymbol{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}$; although the composite function $\text{Tr}(\boldsymbol{J}^{-1})$ is strictly convex on $\boldsymbol{J} \succ \boldsymbol{0}$, so that (P1-C3) itself defines a convex set, the constraint involves the nonlinear matrix inverse and must be transformed into an LMI via the Schur complement before standard SDP solvers can handle it. The binary AP–target association indicators b_{mp} in (P1-C7') can be characterized on the relaxed box by $g(b) = b - b^2 = 0$; this characterization is used later as a concave penalty since $g(b) \geq 0$ for $b \in [0, 1]$ and g is strictly concave. The association gate (P1-C4'a) and the per-AP hardware ceiling (P1-C4'b) are both linear: the left-hand sides are the linear trace terms $\text{Tr}(\boldsymbol{E}_m \boldsymbol{S}_p)$ and $\text{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X)$ extracted by the diagonal selection matrix $\boldsymbol{E}_m$, and the right-hand sides are $P_{\max}\, b_{mp}$ (association-coupled) and $P_{\max}$ (standard hardware upper bound) respectively; both constraints are mixed-integer affine in (P1) and become jointly affine after the box relaxation $0 \leq b_{mp} \leq 1$ used in (P2). The PSD cone $\boldsymbol{S}_p \succeq \boldsymbol{0}$ itself ((P1-C6)), the linear sensing SINR $\text{Tr}(\boldsymbol{g}_p \boldsymbol{g}_p^H \boldsymbol{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2$ ((P1-C2)), and the binary association constraint (P1-C7') is discrete, but because of the four non-convex/nonlinear sources above the overall problem remains a non-convex MINLP. In §4.2 we address each source by a dedicated step of the convexification chain (the rank-1 constraint that arises after covariance lifting is treated as a fifth source and is recovered jointly with the binarity of b_{mp} by the dual DC penalty SCA framework of §4.3; the binary condition belongs to (P1), while its box relaxation and DC penalty are introduced only after lifting in §4.2-E).

4.2 Solution of the Optimization Problem

This section solves the problem along the five non-convex/nonlinear sources identified at the end of §4.1. The SINR fraction and the semi-infinite uncertainty are converted into LMIs via cross-multiplication (§4.2-C step 1) followed by the S-Procedure (§4.2-C step 2); the trace of the PCRB matrix inverse is losslessly converted into an LMI by the Schur complement (§4.2-D); the binarity of the AP–target soft-assignment weights b_{mp} is handled by continuous relaxation plus a DC penalty added to the objective (§4.2-E), which shares a unified dual-DC-penalty SCA framework with the rank-1 constraint in §4.3.

A. Problem Transformation (P2)

To prepare for the semidefinite relaxation, we lift the vector variables in (P1) into covariance matrices, i.e. we define $\boldsymbol{W}_k = \boldsymbol{w}_k \boldsymbol{w}_k^H \in \mathbb{H}_+^{MN_t}$ and $\boldsymbol{R}_X \triangleq \sum_{k=1}^K \boldsymbol{W}_k + \sum_{p=1}^P \boldsymbol{S}_p$, where the target-specific sensing covariances $\boldsymbol{S}_p \in \mathbb{H}_+^{MN_t}$ were introduced in §4.1, yielding (P2):

$$
\begin{aligned}
\min_{\substack{\{\boldsymbol{W}_k\}_{k\in\mathcal{K}},\{\boldsymbol{S}_p\}_{p\in\mathcal{P}},\\ \{b_{mp}\}_{m\in\mathcal{M},p\in\mathcal{P}}}} \quad & \sum_{k=1}^{K} \mathrm{Tr}(\boldsymbol{W}_k) + \sum_{p=1}^{P} \mathrm{Tr}(\boldsymbol{S}_p) && \text{(P2)}\\
\text{s.t.} \quad & \mathrm{Tr}\left(\hat{\boldsymbol{H}}_k(\Delta\boldsymbol{h}_k)\boldsymbol{W}_k\right) - \gamma_k \sum_{j\neq k} \mathrm{Tr}\left(\hat{\boldsymbol{H}}_k(\Delta\boldsymbol{h}_k)\boldsymbol{W}_j\right) \\
& \quad - \gamma_k \,\mathrm{Tr}\Big(\hat{\boldsymbol{H}}_k(\Delta\boldsymbol{h}_k) \textstyle\sum_p \boldsymbol{S}_p\Big) \geq \gamma_k \sigma_c^2, \quad \forall \|\Delta\boldsymbol{h}_k\| \leq \epsilon_h \|\hat{\boldsymbol{h}}_k\|,\ \forall k, \\
& && \text{(P2-C1)}\\
& \mathrm{Tr}(\boldsymbol{g}_p \boldsymbol{g}_p^H \boldsymbol{S}_p) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, && \text{(P2-C2)}\\
& \mathrm{Tr}\left((\boldsymbol{J}_p^{\mathrm{data}}(\boldsymbol{S}_p))^{-1}\right) \leq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, && \text{(P2-C3)}\\
& \mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{S}_p) \leq P_{\max}\, b_{mp}, \quad \forall m \in \mathcal{M},\ p \in \mathcal{P}, && \text{(P2-C4'a)}\\
& \mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}, && \text{(P2-C4'b)}\\
& \sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}}, && \text{(P2-C5')}\\
& \boldsymbol{W}_k \succeq \boldsymbol{0}, \quad \forall k \in \mathcal{K}, && \text{(P2-C6)}\\
& \boldsymbol{S}_p \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}, && \text{(P2-C7)}\\
& \mathrm{rank}(\boldsymbol{W}_k) = 1, \quad \forall k \in \mathcal{K}, && \text{(P2-C8)}\\
& 0 \leq b_{mp} \leq 1, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. && \text{(P2-C9)}
\end{aligned}
$$

Here $\boldsymbol{J}_p^{\text{data}}(\boldsymbol{S}_p) = \frac{2}{\sigma_s^2}\,\text{Re}\{\boldsymbol{D}_p^H \boldsymbol{S}_p \boldsymbol{D}_p\} \in \mathbb{H}_+^{N_\theta \times N_\theta}$ is the data-dependent Fisher Information Matrix under the equivalent MISO sensing observation model, where $\boldsymbol{D}_p = \partial \boldsymbol{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{MN_t \times N_\theta}$ is the effective target-response derivative; $\hat{\boldsymbol{H}}_k(\Delta \boldsymbol{h}_k) \triangleq (\hat{\boldsymbol{h}}_k + \Delta \boldsymbol{h}_k)(\hat{\boldsymbol{h}}_k + \Delta \boldsymbol{h}_k)^H$. This model uses the dedicated sensing covariance $\boldsymbol{S}_p$ directly and does not posit a vectorized multi-receiver architecture. Note that each $\boldsymbol{S}_p$ in (P2) is a joint stacked covariance, so the dedicated waveform for target p may still be synthesized cooperatively across APs; the association gate (P2-C4'a) only authorizes which APs may allocate power to $\boldsymbol{S}_p$, while (P2-C2) sensing SINR and (P2-C3) PCRB are both evaluated from $\boldsymbol{S}_p$ alone.

(P2) retains the exact covariance lifting of the beamforming variables: when $\text{rank}(\boldsymbol{W}_k) = 1$, the mapping $\boldsymbol{w}_k \leftrightarrow \boldsymbol{W}_k = \boldsymbol{w}_k \boldsymbol{w}_k^H$ is recoverable and the quadratic terms are rewritten through identities such as $\text{Tr}(\boldsymbol{x}\boldsymbol{x}^H \boldsymbol{W}) = \boldsymbol{x}^H \boldsymbol{W} \boldsymbol{x}$. However, replacing the binary constraint (P1-C7') by the box constraint (P2-C9) makes (P2) a continuous relaxation of (P1), not an equivalent reformulation. The relaxation can provide a lower bound, while the DC penalty and the subsequent physical-recovery step are used to recover and validate a binary design.

Problem (P2) is the lifted continuous relaxation of (P1). The robust SINR fraction, semi-infinite uncertainty, and PCRB matrix inverse remain to be transformed. The binary association constraint is relaxed to the box (P2-C9) and is recovered later by the DC penalty and physical-validation procedure. The rank-one condition (P2-C8) preserves recoverability of the lifted beamforming covariance but is itself non-convex; SDR subsequently drops this condition and therefore gives a further lower-bound relaxation. The dedicated sensing gate (P2-C4'a), total AP-power limit (P2-C4'b), PSD constraints, and sensing-SINR constraint are affine or convex in the lifted continuous variables.

B. Semi-Definite Relaxation

The rank-1 constraint (P2-C8) in (P2) is addressed by SDR: the rank-1 manifold $\mathcal{F}_{\text{rank-1}}$ is itself non-convex, and SDR relaxes (P2-C8) to $\boldsymbol{W}_k \succeq \boldsymbol{0}$ only, i.e.

$$\boldsymbol{W}_k \succeq \boldsymbol{0}, \quad \forall k, \tag{S1.1}$$

The feasible region is enlarged from $\mathcal{F}_{\text{rank-1}}$ to the positive semidefinite cone $\mathcal{F}_{\text{SDR}}$, and the SDR solution provides a lower bound on the original problem, $P_{\text{SDR}}^* \leq P_{\text{P2}}^*$. For the joint structure of per-user SINR + cell-free cooperation + robust uncertainty, neither the multi-group multicasting tightness condition $G_k \leq 2$ [40] nor the robust MISO condition $N_t \leq 2$ [40] directly applies,

and Gaussian randomization must be used to recover a feasible rank-1 solution when $\mathrm{rank}(\boldsymbol{W}_k^*) > 1$. Specifically, $L \in [100, 1000]$ Gaussian random vectors are generated and the candidate that satisfies the constraints and is optimal in the objective is selected; however, this post-processing has no tight performance-loss upper bound.

C. S-Procedure

This subsection handles the residual semi-infinite robust uncertainty in (P2-C1) in two steps: first the original SINR robust inequality is cross-multiplied into a quadratic form (an equivalent rewrite of (P2-C1), no longer a fraction), and then the S-Procedure reduces the universal quantifier over $\Delta\boldsymbol{h}_k$ to an LMI in the S-Procedure relaxation variable μ_k.

Step 1 – Cross-multiplication. Fix any worst-case $\Delta\boldsymbol{h}_k$ satisfying $\|\Delta\boldsymbol{h}_k\| \le \epsilon_h\|\hat{\boldsymbol{h}}_k\|$. Because $\sigma_c^2 > 0$ guarantees that the denominator is strictly positive, the SINR inequality can be cross-multiplied directly into the quadratic form

$$\mathrm{Tr}(\hat{\boldsymbol{H}}_k\boldsymbol{W}_k) - \gamma_k \sum_{j\neq k} \mathrm{Tr}(\hat{\boldsymbol{H}}_k\boldsymbol{W}_j) - \gamma_k \,\mathrm{Tr}\left(\hat{\boldsymbol{H}}_k \sum_p \boldsymbol{S}_p\right) \ge \gamma_k\sigma_c^2, \qquad \text{(S2.1)}$$

which is a strict equivalence rather than an approximation; strict positivity of the denominator guarantees that the implication holds in both directions. Since σ_c^2 contains no decision variable, the fraction can be eliminated without a first-order Taylor approximation, thus avoiding the accumulation of SCA iteration error.

Step 2 – S-Procedure. Restoring $\Delta\boldsymbol{h}_k$ as a variable and defining $\boldsymbol{A}_k \triangleq \frac{1}{\gamma_k}\boldsymbol{W}_k - \sum_{j\neq k}\boldsymbol{W}_j - \sum_p \boldsymbol{S}_p$, (P2-C1) is rewritten as a quadratic inequality in $\Delta\boldsymbol{h}_k$:

$$\tilde{f}_1(\Delta\boldsymbol{h}) = (\hat{\boldsymbol{h}} + \Delta\boldsymbol{h})^H\boldsymbol{A}_k(\hat{\boldsymbol{h}} + \Delta\boldsymbol{h}) - \sigma_c^2 \ge 0, \quad \forall\|\Delta\boldsymbol{h}_k\| \le \epsilon_h\|\hat{\boldsymbol{h}}_k\|,$$

and the uncertainty set is described by $\tilde{f}_2(\Delta\boldsymbol{h}) = \epsilon_h^2\|\hat{\boldsymbol{h}}_k\|^2 - \|\Delta\boldsymbol{h}_k\|^2 \ge 0$. For this single quadratic constraint on a norm ball, the universal quantifier admits, via the S-Procedure [39], a finite LMI equivalent condition:

Lemma 4.1. ***(S-Procedure).*** *Let* $\tilde{f}_m(\Delta\boldsymbol{h}) = (\Delta\boldsymbol{h})^H\boldsymbol{A}_m(\Delta\boldsymbol{h}) + 2\,\mathrm{Re}\{(\Delta\boldsymbol{h})^H\boldsymbol{b}_m\} + c_m$, $m \in \{1, 2\}$. *Then the implication "*$\tilde{f}_1(\Delta\boldsymbol{h}) \ge 0 \Rightarrow \tilde{f}_2(\Delta\boldsymbol{h}) \ge 0$*" holds* if and only if *there exists* $\mu \ge 0$ *such that*

$$\begin{bmatrix} \boldsymbol{A}_2 - \mu\boldsymbol{A}_1 & \boldsymbol{b}_2 - \mu\boldsymbol{b}_1 \\ \boldsymbol{b}_2^H - \mu\boldsymbol{b}_1^H & c_2 - \mu c_1 \end{bmatrix} \succeq \boldsymbol{0},$$

For a single norm-ball antecedent satisfying Slater's condition, this certificate is lossless. In the application below, the uncertainty function is the antecedent and the SINR quadratic is the consequent.

Applying Lemma 4.1 to the implication $\tilde{f}_2 \geq 0 \Rightarrow \tilde{f}_1 \geq 0$, where $\tilde{f}_2$ describes the norm-ball uncertainty set and $\tilde{f}_1$ is the required SINR quadratic, the worst-case robust form of (P2-C1) can be written as the following LMI:

$$\exists \mu_k \geq 0 : \begin{bmatrix} \boldsymbol{A}_k + \mu_k \boldsymbol{I} & \boldsymbol{A}_k \hat{\boldsymbol{h}}_k \\ \hat{\boldsymbol{h}}_k^H \boldsymbol{A}_k & \hat{\boldsymbol{h}}_k^H \boldsymbol{A}_k \hat{\boldsymbol{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\boldsymbol{h}}_k\|^2 \end{bmatrix} \succeq \boldsymbol{0}. \tag{S3.1}$$

The newly introduced $\mu_k \geq 0$ is the S-Procedure relaxation variable.

D. PCRB Constraint: Schur Complement LMI Transformation

The PCRB constraint (P1-C3) remains in the lifted domain as

$$\mathrm{Tr}\left((\boldsymbol{J}_p^{\mathrm{data}})^{-1}\right) \leq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}, \tag{P2-C3}$$

where $\boldsymbol{J}_p^{\mathrm{data}} \succ \boldsymbol{0}$ is an affine function of $\boldsymbol{S}_p$. The matrix-inverse map $\boldsymbol{J} \mapsto \boldsymbol{J}^{-1}$ is strictly matrix-convex on the positive-definite cone, and the trace operator $\mathrm{Tr}(\cdot)$ is a linear order-preserving map; therefore the composite $\mathrm{Tr}(\boldsymbol{J}^{-1})$ is a strictly convex function and the constraint (P2-C3) itself defines a convex set. However, the constraint involves the nonlinear matrix inverse; to make it directly processable by an interior-point SDP solver, this subsection losslessly converts it into an LMI via the Schur complement.

Introduce an auxiliary Hermitian matrix variable $\boldsymbol{M}_p \in \mathbb{H}^{N_\theta \times N_\theta}$ (where N_θ is the dimension of the target parameter to be estimated). The original constraint can be split equivalently as

$$\boldsymbol{M}_p - (\boldsymbol{J}_p^{\mathrm{data}})^{-1} \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}, \tag{4.5a}$$

$$\mathrm{Tr}(\boldsymbol{M}_p) \leq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}. \tag{4.5b}$$

Since $\boldsymbol{J}_p^{\mathrm{data}} \succ \boldsymbol{0}$, by the Schur complement lemma, (4.5a) is equivalent to

$$\begin{bmatrix} \boldsymbol{M}_p & \boldsymbol{I} \\ \boldsymbol{I} & \boldsymbol{J}_p^{\mathrm{data}} \end{bmatrix} \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}. \tag{4.6}$$

Therefore, the PCRB constraint (P2-C3) is converted completely and without approximation into the linear matrix inequalities in $(\{\boldsymbol{S}_p\}, \{\boldsymbol{M}_p\})$:

$$\begin{bmatrix} \boldsymbol{M}_p & \boldsymbol{I} \\ \boldsymbol{I} & \boldsymbol{J}_p^{\mathrm{data}} \end{bmatrix} \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}, \tag{4.7a}$$

$$\mathrm{Tr}(\boldsymbol{M}_p) \leq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}. \tag{4.7b}$$

This transformation introduces no relaxation gap and avoids outer SCA iterations.

E. AP–Target Soft Assignment: Continuous Relaxation with DC Penalty

This subsection relaxes the original binary condition (P1-C7') to the box constraint (P2-C9) and represents binarity on that box through a DC deficiency function. A path-loss ranking can be used only as a cold-start aid; it is not an optimality claim or a replacement for binary recovery. The DC representation is

$$0 \le b_{mp} \le 1, \tag{E.1a}$$

$$g(b_{mp}) \triangleq b_{mp} - b_{mp}^2 = 0. \tag{E.1b}$$

For $b \in [0,1]$, $g(b) = b - b^2 \ge 0$ holds and g is a concave function of b (since $g''(b) = -2 < 0$); reformulating the equality constraint (E.1b) as a penalty term added to the objective gives

$$\min \quad \underbrace{\sum_{k=1}^{K} \mathrm{Tr}(\boldsymbol{W}_k) + \sum_{p=1}^{P} \mathrm{Tr}(\boldsymbol{S}_p)}_{\text{original objective}} + \eta_b \underbrace{\sum_{m\in\mathcal{M}, p\in\mathcal{P}} g(b_{mp})}_{=\sum_{m,p}(b_{mp}-b_{mp}^2)} \quad \text{s.t. (P2-C1)} \sim \text{(P2-C9)}\backslash\{\text{(P2-C9)}\}, \tag{P2-DCP}$$

where $\eta_b \ge 0$ is the penalty factor; the objective exhibits a DC structure ("convex function + concave function $\times\eta_b$") and the box constraints (P2-C9) are kept in their convex form, but the term $-\eta_b b_{mp}^2$ in the objective is concave in b_{mp}, so (P2-DCP) is not a convex optimization problem and this non-convexity is handled jointly with the rank-1 constraint by the dual DC penalty SCA framework in §4.3. For $b_{mp} \in \{0,1\}$ we have $g(b_{mp}) = 0$ and the penalty incurs no cost on binary solutions; for $b_{mp} \in (0,1)$ we have $g(b_{mp}) > 0$ and the penalty grows with η_b, thus increasing η_b encourages small binary deficiency, but no finite penalty alone guarantees $b_{mp}^\star \in \{0,1\}$. Binary recovery is certified only after the candidate is fixed and physically validated. After introducing the η_b concave penalty, (P2-DCP) has exactly the same mathematical structure as the rank-1 penalty $\rho(\boldsymbol{W}_k) = \mathrm{Tr}(\boldsymbol{W}_k) - \lambda_{\max}(\boldsymbol{W}_k)$ introduced in §4.3-A—both are "convex − convex" DC forms—so the SCA framework designed in §4.3 can carry both penalties simultaneously: within a single SCA iteration we perform an eigendecomposition of $\boldsymbol{W}_k$ and a first-order Taylor expansion of g at b_{mp}, yielding a unified convex SDP subproblem in the variables $(\{\boldsymbol{W}_k\}, \{b_{mp}\}, \{\boldsymbol{S}_p\}, \{\mu_k\}, \{\boldsymbol{M}_p\})$. The box constraint $0 \le$

$b_{mp} \leq 1$ of (P2-DCP) restricts the SCA cold start; in practice we may use a *large-scale-fading heuristic* to generate the initial point—for each target p, sort APs by $\mathrm{PL}(d_{m,p})$ in ascending order, set $b_{mp}^{(0)} = 1$ for the top-N_{req} APs and $b_{mp}^{(0)} = 0$ for the rest; this value automatically satisfies the box constraint and the equality (P2-C5'), and is at most one step away from the $\{0, 1\}$ boundary. *This heuristic serves only as a cold-start aid for SCA and is no longer the final solver*—the final candidate can depend on the SCA initialization and candidate-recovery path; physical validation is therefore an essential final step.

F. Consolidation (P3): Final Form with Dual DC Penalties

Applying the SDR step of §-B ((S1.1)), the S-Procedure step of §-C ((S2.1) + (S3.1)), the Schur complement of §-D ((4.7)), and the continuous relaxation + DC penalty of §-E (box (P2-C9) + (P2-DCP)) to (P2), we obtain the final problem (P3):

$$\min_{\substack{\{\boldsymbol{W}_k\},\{\boldsymbol{S}_p\},\{\mu_k\},\\ \{\boldsymbol{M}_p\},\{b_{mp}\}}} \quad \sum_{k=1}^{K} \mathrm{Tr}(\boldsymbol{W}_k) + \sum_{p=1}^{P} \mathrm{Tr}(\boldsymbol{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2) \tag{P3}$$

$$\text{s.t.} \quad \begin{bmatrix} \boldsymbol{A}_k + \mu_k \boldsymbol{I} & \boldsymbol{A}_k \hat{\boldsymbol{h}}_k \\ \hat{\boldsymbol{h}}_k^H \boldsymbol{A}_k & \hat{\boldsymbol{h}}_k^H \boldsymbol{A}_k \hat{\boldsymbol{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\boldsymbol{h}}_k\|^2 \end{bmatrix} \succeq \boldsymbol{0}, \quad \forall k \tag{P3-C1}$$

$$\mathrm{Tr}(\boldsymbol{g}_p \boldsymbol{g}_p^H \boldsymbol{S}_p) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{P3-C2}$$

$$\begin{bmatrix} \boldsymbol{M}_p & \boldsymbol{I} \\ \boldsymbol{I} & \boldsymbol{J}_p^{\mathrm{data}} \end{bmatrix} \succeq \boldsymbol{0}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{P3-C3}$$

$$\mathrm{Tr}(\boldsymbol{M}_p) \leq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{P3-C3b}$$

$$\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{S}_p) \leq P_{\max}\, b_{mp}, \quad \forall m, p \tag{P3-C4'a}$$

$$\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X) \leq P_{\max}, \quad \forall m \tag{P3-C4'b}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{P3-C5'}$$

$$\boldsymbol{W}_k \succeq \boldsymbol{0}, \quad \forall k \tag{P3-C6}$$

$$\boldsymbol{S}_p \succeq \boldsymbol{0}, \quad \forall p \tag{P3-C7}$$

$$\mu_k \geq 0, \quad \forall k \tag{P3-C8}$$

$$0 \leq b_{mp} \leq 1, \quad \forall m, p. \tag{P3-C9}$$

where $\boldsymbol{A}_k = \frac{1}{\gamma_k}\boldsymbol{W}_k - \sum_{j\neq k}\boldsymbol{W}_j - \sum_p \boldsymbol{S}_p$, $\boldsymbol{J}_p^{\text{data}}$ is an affine function of $\boldsymbol{S}_p$, and $\boldsymbol{E}_m$ is the antenna-block-extraction diagonal matrix. The feasible region of (P3) is the intersection of the LMI set (P3-C1), the LMI set (P3-C3), the linear equalities/inequalities (P3-C2)/(P3-C3b)/(P3-C4'a)/(P3-C4'b)/(P3-C5'), the positive-semidefinite cones (P3-C6)/(P3-C7), the scalar half-space (P3-C8), and the box constraints (P3-C9); each of these sets is convex, so the feasible region is convex. The objective $\sum_k \text{Tr}(\boldsymbol{W}_k)+\sum_p \text{Tr}(\boldsymbol{S}_p)+\eta_b \sum_{m,p}(b_{mp}-b_{mp}^2)$ is linear in $(\{\boldsymbol{W}_k\},\{\boldsymbol{S}_p\})$ and is a "linear $-\eta_b\cdot$ quadratic" DC form in $\{b_{mp}\}$, and the overall objective is DC (non-convex) in $(\{\boldsymbol{W}_k\},\{\boldsymbol{S}_p\},\{b_{mp}\})$, so (P3) is not a convex SDP and its non-convexity is solved by the dual DC penalty SCA framework of §4.3.

Theorem 4.2 (Structural Properties of (P3))**.** *The feasible region of (P3) is convex; the objective is a DC function of* $(\{\boldsymbol{W}_k\},\{b_{mp}\},\{\boldsymbol{S}_p\})$*; the constraint matrices are affine in* $(\{\boldsymbol{W}_k\},\{\boldsymbol{S}_p\})$ *and linear in* $\{b_{mp}\}$*. Hence (P3) is a difference-of-convex program (DCP), and a stationary point can be obtained by the dual DC penalty SCA framework of §4.3.*

4.3 Joint DC Penalty SCA Framework: Rank-One and Binary Recovery

The feasible region of (P3) is convex but its objective is DC, so obtaining a global optimum directly is hard. A KKT stationary point is a candidate for recovery, not by itself a certificate of a binary rank-one physical design. This section designs a dual DC penalty SCA framework: the concave term $-\eta_b b_{mp}^2$ in the (P3) objective and the concave term $-\eta_{\text{rank}}\lambda_{\max}(\boldsymbol{W}_k)$ (corresponding to the missing rank-1 property) are linearized simultaneously within the same SCA loop, and each iteration solves a standard convex SDP. This section also presents Gaussian randomization / leading-eigenvector extraction as a baseline post-processing.

A. Unified Characterization of the Dual DC Penalties

For any positive-semidefinite matrix $\boldsymbol{W}_k \succeq \boldsymbol{0}$, it has rank one if and only if its trace equals its largest eigenvalue:

$$\text{rank}(\boldsymbol{W}_k) = 1 \quad \Longleftrightarrow \quad \text{Tr}(\boldsymbol{W}_k) - \lambda_{\max}(\boldsymbol{W}_k) = 0. \tag{4.9}$$

Since $\lambda_{\max}(\boldsymbol{W}_k) \leq \text{Tr}(\boldsymbol{W}_k)$ always holds for PSD matrices, define the rank-deficiency measure

$$\rho(\boldsymbol{W}_k) \triangleq \text{Tr}(\boldsymbol{W}_k) - \lambda_{\max}(\boldsymbol{W}_k) \geq 0. \tag{4.10}$$

For $b \in [0,1]$, define the binarity-deficiency measure

$$g(b) \triangleq b - b^2 = b(1-b) \geq 0, \quad g''(b) = -2 < 0 \ \text{(strictly concave)}. \tag{4.11}$$

$\rho(\boldsymbol{W}_k)$ and $g(b_{mp})$ are mathematically isomorphic—both are "convex − convex" DC forms, vanish on the desired constraint set, and are positive when the constraint is violated; this isomorphism is the mathematical foundation that allows the two penalties to share a single SCA framework. Adding the rank-1 penalty $\eta_{\text{rank}} \sum_k \rho(\boldsymbol{W}_k)$ to the (P3) objective gives

$$\begin{aligned} \min \quad & \underbrace{\sum_k \text{Tr}(\boldsymbol{W}_k) + \sum_p \text{Tr}(\boldsymbol{S}_p)}_{\text{original objective}} + \underbrace{\eta_{\text{rank}} \sum_k \rho(\boldsymbol{W}_k)}_{\text{rank-1 penalty (DC)}} + \underbrace{\eta_b \sum_{m,p} g(b_{mp})}_{\text{binarity penalty (DC)}} \\ & \text{s.t. (P3-C1)} \sim \text{(P3-C9)}. \end{aligned} \tag{4.12}$$

All constraints of (P3-DCP) are convex (inherited from the convex feasible region of (P3)), and the only non-convexity comes from the two DC penalty terms in the objective.

B. SCA Linearization: Simultaneous First-Order Taylor Expansion of Both DC Terms

At iteration t, let the current iterate be $(\{\boldsymbol{W}_k^{(t)}\}, \{b_{mp}^{(t)}\})$. Perform the eigen-decomposition

$$\boldsymbol{W}_k^{(t)} = \sum_{i=1}^{r_k} \lambda_{k,i}\, \boldsymbol{u}_{k,i} \boldsymbol{u}_{k,i}^H, \quad \lambda_{k,1} \geq \lambda_{k,2} \geq \cdots \geq \lambda_{k,r_k} > 0, \tag{4.13}$$

where the leading eigenvector is $\boldsymbol{u}_{\max,k}^{(t)} \triangleq \boldsymbol{u}_{k,1}$. By the convexity of $\lambda_{\max}(\boldsymbol{W}_k)$, its first-order Taylor expansion provides a global lower bound:

$$\lambda_{\max}(\boldsymbol{W}_k) \geq \lambda_{\max}(\boldsymbol{W}_k^{(t)}) + \text{Tr}\left(\boldsymbol{u}_{\max,k}^{(t)} (\boldsymbol{u}_{\max,k}^{(t)})^H (\boldsymbol{W}_k - \boldsymbol{W}_k^{(t)})\right) = \text{Tr}\left(\boldsymbol{u}_{\max,k}^{(t)} (\boldsymbol{u}_{\max,k}^{(t)})^H \boldsymbol{W}_k\right). \tag{4.14}$$

Substituting this into the rank-1 penalty of (P3-DCP) yields a global upper bound (the concave term is replaced by its upper bound, the subproblem minimizes, and the true objective is a lower bound of the subproblem optimum):

$$\rho(\boldsymbol{W}_k) = \text{Tr}(\boldsymbol{W}_k) - \lambda_{\max}(\boldsymbol{W}_k) \leq \text{Tr}(\boldsymbol{W}_k) - \text{Tr}\left(\boldsymbol{u}_{\max,k}^{(t)} (\boldsymbol{u}_{\max,k}^{(t)})^H \boldsymbol{W}_k\right) \triangleq \hat{\rho}^{(t)}(\boldsymbol{W}_k). \tag{4.15}$$

The first-order Taylor expansion of the concave $g(b) = b - b^2$ at $b^{(t)}$ gives the global overestimator

$$g(b) \le g(b^{(t)}) + g'(b^{(t)})(b - b^{(t)}) = b^{(t)} - (b^{(t)})^2 + (1 - 2b^{(t)})(b - b^{(t)}), \quad (4.16)$$

and after removing the constant term $b^{(t)} - (b^{(t)})^2 - (1 - 2b^{(t)})b^{(t)} = (b^{(t)})^2$ (which does not affect minimization), the linearized penalty is

$$g(b) \le (1 - 2b^{(t)})\, b + \text{const} \triangleq \hat{g}^{(t)}(b). \quad (4.17)$$

Substituting (4.15) and (4.17) into (P3-DCP) gives the convex SDP subproblem at iteration t:

$$\begin{aligned} \min_{\substack{\{\boldsymbol{W}_k\},\{\boldsymbol{S}_p\},\{\mu_k\},\\ \{\boldsymbol{M}_p\},\{b_{mp}\}}} \quad & \sum_{k=1}^{K} \mathrm{Tr}(\boldsymbol{W}_k) + \sum_{p=1}^{P} \mathrm{Tr}(\boldsymbol{S}_p) + \eta_b \sum_{m,p} (1 - 2b_{mp}^{(t)})\, b_{mp} \\ & + \eta_{\mathrm{rank}} \sum_{k=1}^{K} \Big[\mathrm{Tr}(\boldsymbol{W}_k) - \mathrm{Tr}\left(\boldsymbol{u}_{\max,k}^{(t)} (\boldsymbol{u}_{\max,k}^{(t)})^H \boldsymbol{W}_k\right) \Big] \\ \text{s.t.} \quad & \text{(P3-C1)} \sim \text{(P3-C9)}. \end{aligned} \quad (4.18)$$

The objective of (4.18) is linear in $(\{\boldsymbol{W}_k\}, \{\boldsymbol{S}_p\}, \{b_{mp}\})$ (the η_{rank} term combines $\mathrm{Tr}(\boldsymbol{W}_k)$ with $-\mathrm{Tr}(\boldsymbol{u}_{\max,k}^{(t)}(\boldsymbol{u}_{\max,k}^{(t)})^H \boldsymbol{W}_k)$ to form $\mathrm{Tr}((\boldsymbol{I} - \boldsymbol{u}_{\max,k}^{(t)}(\boldsymbol{u}_{\max,k}^{(t)})^H)\boldsymbol{W}_k)$, and the η_b term is linear in b_{mp}), and the constraints (P3-C1)–(P3-C9) are all LMI / linear / positive-semidefinite / box, so (P3-SCA-t) is a standard convex SDP that can be efficiently solved by interior-point solvers such as MOSEK/SDPT3. Update the iterate $(\{\boldsymbol{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\boldsymbol{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\}, \{\boldsymbol{M}_p^{(t+1)}\})$ and repeat.

Theorem 4.3 (Dual DC SCA Limit-Point Property)**.** *Assume fixed penalty factors and feasible SCA subproblems. The majorization–minimization updates produce a non-increasing sequence of penalized objective values and hence a convergent objective sequence. Finite penalty factors alone do not guarantee exact rank-one or binary recovery. If a limit point satisfies $\rho(\boldsymbol{W}_k^\star) = 0$ for every k and $g(b_{mp}^\star) = 0$ for every (m, p), then the recovered variables satisfy the lifted constraints corresponding to (P1). Direct physical validation of the extracted beamformers and binary associations is still required to certify (P1-C1)–(P1-C7').*

C. Baseline: Gaussian Randomization / Eigen-Extraction

If one pursues the fastest single-SDP run, the master problem (P3) with both DC penalties removed ($\eta_{\mathrm{rank}} = \eta_b = 0$) can be solved directly and the rank-one and binary solutions are recovered by post-processing: $\boldsymbol{W}_k$ is recovered

Algorithm 1 Dual DC Penalty SCA Iteration (Joint Rank-One and Binary Recovery)

Require: Initial feasible point $\{\boldsymbol{W}_k^{(0)}, b_{mp}^{(0)}\}$ (generated by the heuristic of §4.2-E), penalty factors $\eta_{\text{rank}}, \eta_b > 0$, tolerance $\varepsilon > 0$, maximum number of iterations $T_{\max}$.

Ensure: Rank-one (or approximately rank-one) beamforming covariance $\{\boldsymbol{W}_k^\star\}$, binary AP–target assignment $\{b_{mp}^\star \in \{0,1\}\}$, sensing covariances $\{\boldsymbol{S}_p^\star\}$.

Solve (P3) with both DC penalties removed ($\eta_{\text{rank}} = \eta_b = 0$, i.e., the SDR master with box (P3-C9)) to obtain an initial SDR solution; this solution may have $b_{mp} \notin \{0,1\}$ when $\eta_b = 0$, but it is feasible as an SCA starting point. Report infeasible if no such solution exists.
for $t = 0, 1, \ldots, T_{\max} - 1$ **do**
    for each $k \in \mathcal{K}$ **do**
        Eigendecompose $\boldsymbol{W}_k^{(t)}$ to obtain the leading eigenvector $\boldsymbol{u}_{\max,k}^{(t)}$.
    end for
    for each (m,p) **do**
        Compute the g' coefficient $c_{mp}^{(t)} = 1 - 2b_{mp}^{(t)}$.
    end for
    Solve the convex SDP (4.18) for the next iterate $(\{\boldsymbol{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\boldsymbol{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\}, \{\boldsymbol{M}_p^{(t+1)}\})$.
    Compute $\delta_{\text{rank}}^{(t)} = \sum_k \rho(\boldsymbol{W}_k^{(t+1)})$ and $\delta_b^{(t)} = \sum_{m,p} g(b_{mp}^{(t+1)})$.
    if $\delta_{\text{rank}}^{(t)} \le \varepsilon$ and $\delta_b^{(t)} \le \varepsilon$ and the change in objective $|P^{(t+1)} - P^{(t)}| \le \varepsilon$ **then**
        break.
    end if
end for
For each k, set $\boldsymbol{w}_k^\star = \sqrt{\lambda_{\max}(\boldsymbol{W}_k^{(t+1)})}\, \boldsymbol{u}_{\max,k}^{(t+1)}$.
For each (m,p), round $b_{mp}^\star = \mathbb{1}[b_{mp}^{(t+1)} \ge 0.5]$ (if $\delta_b^{(t)}$ is sufficiently small, the rounding incurs no cost).
return $\{\boldsymbol{w}_k^\star, b_{mp}^\star, \{\boldsymbol{S}_p^\star\}\}$.

as $\boldsymbol{w}_k^\star$ via leading-eigenvector extraction and power scaling in Algorithm 2, and b_{mp} is rounded to $\{0, 1\}$ by a threshold of 0.5. It must be emphasized that this kind of post-processing breaks the strict robust-SINR guarantee of the S-Procedure and the exact satisfaction of the box constraints, and additional power scaling may be required in practice to restore feasibility.

Algorithm 2 Rank-One Beamforming Extraction (Baseline Post-Processing)

Require: Converged solution $\{\boldsymbol{W}_k^\star, \{\boldsymbol{S}_p^\star\}, b_{mp}^\star\}$.
Ensure: Rank-one beamforming vectors $\{\boldsymbol{w}_k^\star\}$.

for each $k \in \mathcal{K}$ **do**
    Compute the eigendecomposition $\boldsymbol{W}_k^\star = \boldsymbol{U}_k \boldsymbol{\Lambda}_k \boldsymbol{U}_k^H$.
    Extract the leading eigenvector: $\tilde{\boldsymbol{w}}_k = \sqrt{\lambda_{\max}(\boldsymbol{\Lambda}_k)} \cdot \boldsymbol{u}_k^{\max}$.
    Scale to satisfy the per-AP power: $\boldsymbol{w}_k^\star = \eta_k \tilde{\boldsymbol{w}}_k$, where η_k is chosen so that $\mathrm{Tr}(\boldsymbol{E}_m \boldsymbol{R}_X) \leq P_{\max}$ with $\boldsymbol{R}_X = \sum_k \boldsymbol{w}_k^\star (\boldsymbol{w}_k^\star)^H + \sum_p \boldsymbol{S}_p^\star$ for every m.
end for
if $\{\boldsymbol{w}_k^\star, b_{mp}^\star\}$ satisfies all of (P1-C1)–(P1-C7') **then**
    return $\{\boldsymbol{w}_k^\star, b_{mp}^\star\}$.
else
    Apply a power-scaling fallback: $\boldsymbol{w}_k^\star \leftarrow \alpha \boldsymbol{w}_k^\star$, with $\alpha \in (0, 1]$ chosen to restore feasibility.
    return $\{\boldsymbol{w}_k^\star, b_{mp}^\star\}$.
end if

D. Proposed Workflow: Candidate Re-Solution and Physical Validation

Algorithms 1 and 2 produce relaxed or post-processed candidates. The proposed workflow evaluated in Chapter 5 adds a decisive final stage: each binary candidate topology is fixed, the remaining continuous problem is re-solved, and only physically validated designs are compared. Candidate topologies comprise the rounded DC-SCA assignment (top-N_{req} by association weight), a distance-based path-loss heuristic, and two user-aware variants; duplicate topologies are removed before re-solution.

Algorithm 3 Topology-Aware Recovery: Fixed-Topology Re-Solution and Physical Validation (Proposed)

Require: SCA output $\{\boldsymbol{W}_k^{(\cdot)}, b_{mp}^{(\cdot)}, \boldsymbol{S}_p^{(\cdot)}\}$ from Algorithm 1; validation tolerances.

Ensure: Physically validated design $\{\boldsymbol{w}_k^\star, b_{mp}^\star, \{\boldsymbol{S}_p^\star\}\}$, or an infeasibility report.

Form binary candidates: the rounded DC assignment, the distance/path-loss top-N_{req} heuristic, and the two user-aware variants; discard duplicates.
for each candidate topology $\hat{b}$ **do**
    Fix $b_{mp} = \hat{b}_{mp}$ and re-solve (P3) over the continuous variables only.
    Extract $\boldsymbol{w}_k$ by leading-eigenvector extraction with power scaling.
    Directly validate (P1-C1)–(P1-C7'): worst-case communication SINR, sensing SINR, matrix PCRB, per-AP power, and cardinality.
    if the recovered design passes all checks **then**
        Record it as a feasible candidate with its physical objective value.
    end if
end for
return the feasible candidate with the lowest recovered transmit power; if none is feasible, report infeasible.

4.4 Computational Complexity and Scalability

The proposed procedure is polynomial per convex subproblem but is not a single-SDP method. Its cost has two distinct components: the cost of solving one dense SCA SDP and the number of such solves induced by candidate recovery. This section therefore reports a dimension-level characterization rather than an implementation-independent exact flop count; the latter depends on the interior-point solver, sparsity exploitation, stopping tolerances, and warm-start behavior.

Let $N = MN_t$ be the stacked transmit dimension. In one call to `solve_p3_sca_t`, the K communication covariances and the P target-specific sensing covariances contribute $(K+P)N^2$ real-order Hermitian degrees of freedom. The remaining variables contribute MP relaxed association variables, K S-Procedure multipliers, and PN_θ^2 PCRB auxiliary-matrix entries. Thus the scalar decision dimension is of order

$$n_{\text{var}} = \mathcal{O}\big((K+P)N^2 + MP + K + PN_\theta^2\big)\,. \tag{4.19}$$

The conic representation contains K robust-SINR cones of order $N+1$, P PCRB cones of order $2N_\theta$ (for $N_\theta > 1$), and $K+P$ PSD covariance cones of order N, in addition to scalar affine constraints. A dense primal–dual interior-point method must at least form and factor quantities whose cubic cone-size contribution scales as

$$\mathcal{O}\big(K(N+1)^3 + (K+P)N^3 + P(2N_\theta)^3\big) \tag{4.20}$$

per Newton step, plus the solver-dependent Schur-complement linear-system cost associated with n_{var}. Consequently, the dominant dependence grows faster than linearly with both AP count M and antennas per AP N_t, because both increase $N = MN_t$ and hence the covariance and LMI dimensions. Each SCA iteration also performs K leading-eigenvector computations, adding $\mathcal{O}(KN^3)$ dense linear-algebra work; this is normally secondary to the SDP solve at the tested dimensions.

The recovery loop multiplies, but does not exponentially enumerate, this per-SDP cost. In the implementation of Algorithm 3, one unpenalized SDR gatekeeper is solved first. Two initial association patterns then run at most five dual-DC iterations each. Their rounded outputs and the three deterministic topology rules yield at most five distinct binary candidates after duplicate removal. A fixed-topology re-solve uses one unpenalized initialization and at most ten rank-DC iterations. Hence, before early stopping, the

implementation has the explicit upper bound

$$N_{\mathrm{SDP}} \leq 1 + 2 \times 5 + 5 \times (1 + 10) = 66 \tag{4.21}$$

SDP calls per scenario. In typical runs duplicate topologies and convergence checks reduce this number. Importantly, this is a bounded candidate-recovery schedule, not an exhaustive search over the $\binom{M}{N_{\mathrm{req}}}^{P}$ possible authorization patterns.

The complexity statement explains the empirical timing results in Chapter 5: increasing infrastructure improves physical feasibility and QoS margins, but the growing $N \times N$ covariance cones and the repeated fixed-topology re-solves increase wall-clock time superlinearly. The method is therefore appropriate for offline planning or slow-timescale resource updates at the tested dimensions. Distributed decomposition, candidate pruning, and warm-starting are the natural routes for scaling the same physical-validation workflow further, as discussed in Chapter 6.

4.5 Link to Physical Evaluation

The convexification chain represents the robust SINR and PCRB conditions by LMIs in the lifted variables, while the dual DC procedure drives the rank and binary deficiencies toward zero. Neither a finite penalty nor a stationary point alone proves exact recovery or global optimality. The final step is therefore candidate-specific: recover or re-solve the binary topology, extract the beamformers, and directly validate (P1-C1)–(P1-C7'); this is exactly the workflow of Algorithm 3, which is what 'proposed full recovery" refers to in Chapter 5. Chapter 5 uses this distinction to compare methods by physical feasibility first and conditional performance second.

Chapter 5

Numerical Evaluation and Results Analysis

This chapter tests the three links in the thesis contribution chain: whether topology-aware recovery improves physically realizable sensing designs, whether the robust formulation protects communication QoS under CSI uncertainty, and whether the complete workflow remains useful after recovery and at a larger network dimension. Each result is interpreted with its feasibility evidence, not only with a conditional performance average.

5.1 Evaluation Protocol, Metrics, and Baselines

The experiments use the result sets stored with the paper figures. A run is counted as physically feasible only after binary association recovery, rank-one beamformer extraction or fixed-topology re-solution, and direct validation of the recovered design. The validation checks the robust communication SINR, sensing SINR, matrix PCRB, per-AP transmit-power limit, and association cardinality. Consequently, an SDR solution or a continuous association vector is not reported as a feasible physical design by itself.

All compared methods use common random scenarios within a sweep. Feasibility rates are therefore unconditional statistics over those scenarios. Power, runtime, rate, sensing-SINR, and PCRB summaries marked "conditional" are computed only over the recovered designs that pass physical validation. The baselines separate different design choices: FIM-greedy and nearest-AP methods fix topology according to sensing geometry, random topology removes geometry awareness, and relaxation or single-penalty variants isolate the recovery mechanism. This protocol ensures that a low con-

ditional power is never read as an advantage when it is accompanied by frequent infeasibility.

Two dispersion conventions are used consistently in this chapter. Where the central statistic is a conditional mean, as in Figure 5.1, the shaded band spans ± 1 standard deviation across the feasible-recovery seeds. Where the central statistic is a median, as in Figure 5.5, the shaded band spans the interquartile range (25th–75th percentiles) across seeds. Both bands describe scenario-to-scenario variability of the recovered designs; they are not confidence intervals on an estimated mean.

Table 5.1 records the common reference settings and the deliberate deviations used by the result families. It is not a claim that every figure uses one universal network size: the purpose of the table is to make the reference point and every reported sweep dimension explicit. In particular, N_{req} is configuration-specific because it is the quantity swept in Figure 5.1 and is fixed separately in the larger-scale test.

5.2 Topology-Recovery Gains in Ordinary and Stress Geometries

Figure 5.1 compares topology-recovery strategies over 30 common seeds for a six-AP configuration. The proposed full recovery and the FIM-greedy topology are physically feasible in all displayed cluster-size settings. In contrast, the nearest-AP baseline is infeasible in 20% of cases at $N_{\mathrm{req}} = 2$, and the random-topology baseline is substantially less reliable for small and intermediate cluster sizes. The comparison shows that the recovery workflow can identify physically usable sensing support where purely local or random topology rules fail; the comparison is not evidence that relaxation alone is exact.

The cardinality equality in (P1-C5') is a resource-allocation requirement, not a monotone performance knob. Increasing N_{req} changes the authorized sensing geometry and the feasible power allocation simultaneously, so conditional power need not vary monotonically. The PCRB ratios remain close to one in feasible recovered cases, which is consistent with the tracking constraint being active at the solution.

Figure 5.2 asks whether the same topology advantage persists when the geometry is deliberately unfavorable. With colocated edge users and targets, and again with crowded targets, the proposed full recovery and FIM-greedy topology remain feasible in all shown cases. The nearest-AP topology loses feasibility in the edge-colocated setting (29/30 feasible) and falls to about

Table 5.1: Reference simulation settings and sweep ranges. A reference value applies only when the corresponding quantity is not the sweep variable of a figure.

Item	Reference setting or reported range
Reference factor-sweep network	$M = 6$, $N_t = 2$, $K = 3$, $P = 2$; $N_\theta = 2$; square area side length 400 m.
AP and load sweeps	$M \in \{4, 6, 8, 10\}$, $N_t \in \{1, 2, 4\}$, $K \in \{2, 3, 4, 5\}$, and $P \in \{1, 2, 3\}$.
Area and power sweeps	Area side length $\{200, 400, 600\}$ m; per-AP budget $P_{\max} \in \{17, 20, 23\}$ dBm (reference 20 dBm).
Channel and noise model	Rayleigh communication channels and LoS/Rician target responses at 2.8 GHz; normalized $\sigma_c^2 = \sigma_s^2 = 1$.
QoS and uncertainty	Communication and sensing SINR targets of 0 dB; relative communication-CSI radius $\epsilon_h = 0.05$ except in the CSI-robustness sweep. The PCRB threshold is calibrated per target from a feasible isotropic reference.
Sensing authorization	$\sum_m b_{mp} = N_{\text{req}}$. N_{req} is swept in Figure 5.1, while the larger-scale validation uses $N_{\text{req}} = 3$; other figures use their recorded configuration-specific value.
Monte-Carlo protocol	Thirty common seeds for each extended-factor configuration; 25 common seeds for the dual-DC ablation; eight common seeds for the $M = 12$ validation.

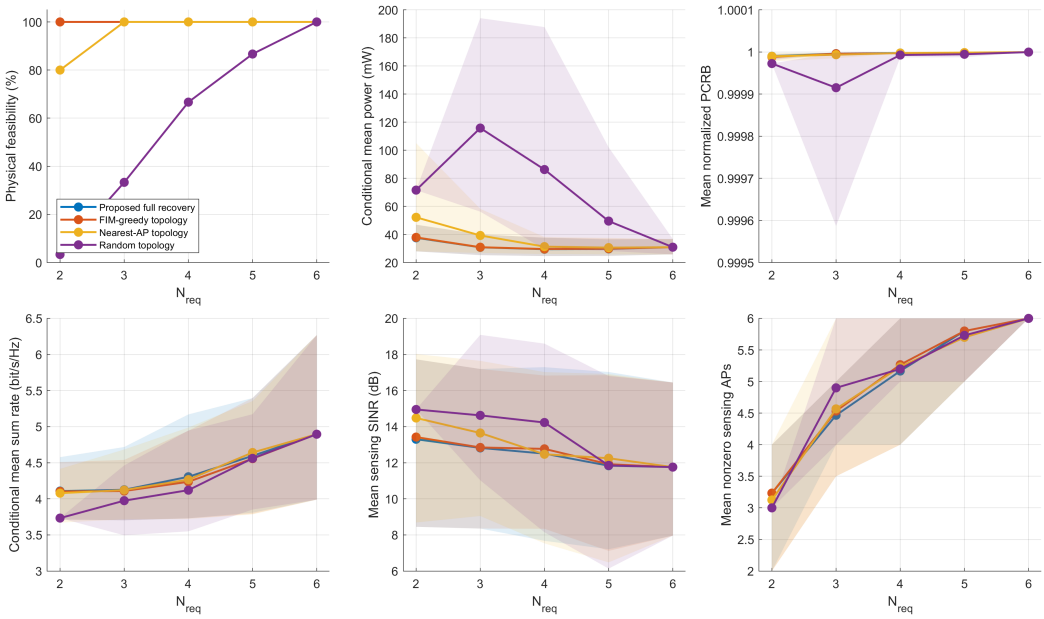


Figure 5.1: Topology-method comparison versus required sensing APs per target. Feasibility is measured over all common scenarios; power, rate, sensing-SINR, and PCRB summaries are conditional on physically feasible recovered designs. Shaded bands span ± 1 standard deviation across the feasible-recovery seeds.

23% in the crowded setting. When it succeeds in the crowded case, its conditional power approaches 300 mW, whereas the proposed and FIM-greedy methods remain near 50 mW. The physical mechanism is geometric: selecting the nearest APs alone can exhaust useful sensing directions or AP power under crowding. The evidence is limited to the tested pressure configurations, but it demonstrates why feasibility must accompany every conditional comparison.

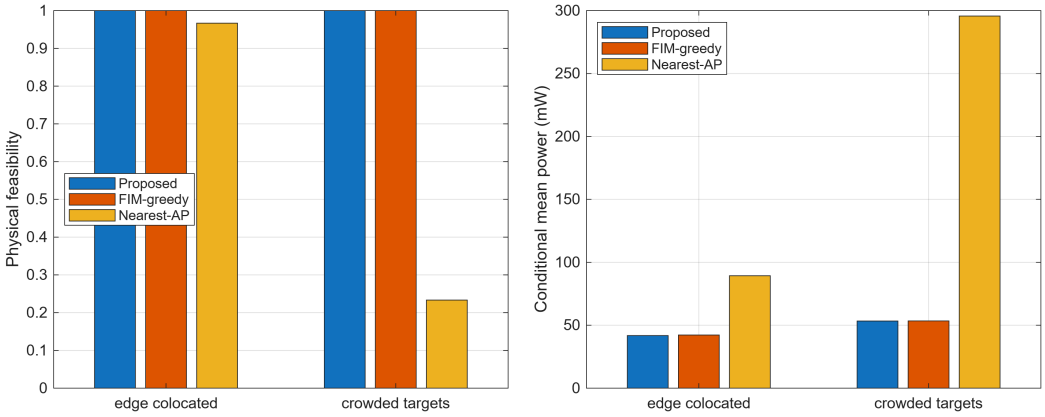


Figure 5.2: Physical feasibility and conditional power in two pressure geometries. Conditional-power bars exclude infeasible recovered runs and must be interpreted together with the associated feasibility rate.

5.3 Robustness–Power Trade-Off under CSI Uncertainty

Figure 5.3 compares the robust design with a nominal-CSI baseline as the normalized CSI-uncertainty radius increases. Both methods are certified feasible at their respective design points. When tested against perturbations within the uncertainty ball, the robust design has zero observed outage throughout the displayed sweep, whereas the nominal-CSI design has approximately 90% outage for nonzero uncertainty radii. The robust protection raises transmit power moderately, from about 28.7 mW at zero uncertainty to about 30 mW at the largest displayed radius.

This is the intended robustness–power trade-off: the additional power is associated with markedly improved robustness under the tested perturbations. It should not be interpreted as a guarantee outside the specified norm-bounded uncertainty model, nor as a claim that the nominal design is infeasible at the channel estimate itself.

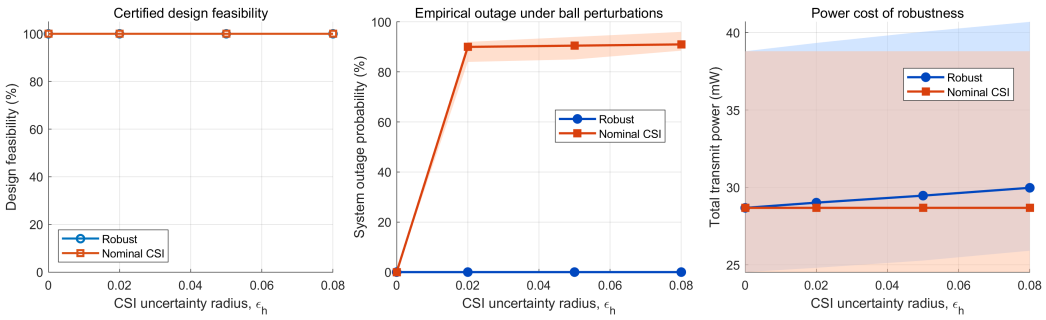


Figure 5.3: Robustness comparison under increasing CSI-uncertainty radius. The middle panel is an empirical perturbation test and the right panel reports the corresponding transmit-power cost.

5.4 Recovery Mechanism and DC-SCA Stabilization

The preceding comparisons concern recovered physical designs. Figure 5.4 therefore isolates the mechanism that makes recovery reliable. Across 25 common scenarios, all four variants have the same observed physical-feasibility rate of 92%, so the ablation does not show that one penalty alone enlarges the feasible set in this test. Its informative result is recovery quality: relaxation and rank-only variants retain a binary distance on the order of 10^{-1}, whereas binary-only and dual-DC variants reduce the median binary distance below the 10^{-4} reference. The dual-DC variant also maintains a very small median rank residual. Thus both penalties are needed when the final design must be binary and rank one simultaneously.

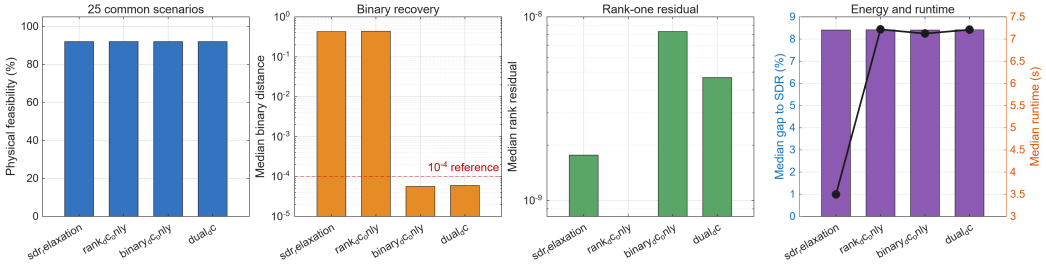


Figure 5.4: Recovery ablation over 25 common scenarios. Physical feasibility is evaluated after recovery; binary distance and rank residual diagnose whether a relaxed solution is suitable for physical recovery.

Figure 5.5 provides the iteration-level complement to the ablation on the same 25 seeds. The median binary distance falls from approximately 0.43 at

initialization to the 10^{-3} scale after the first DC-SCA iteration and below 10^{-4} from the second iteration onward; the recovery-ready fraction remains at 100% over the displayed iterations. The median rank residual stays at the 10^{-9} scale. The continuous-power trace is a continuation diagnostic only; it is not used to assert monotone descent of the original non-convex objective.

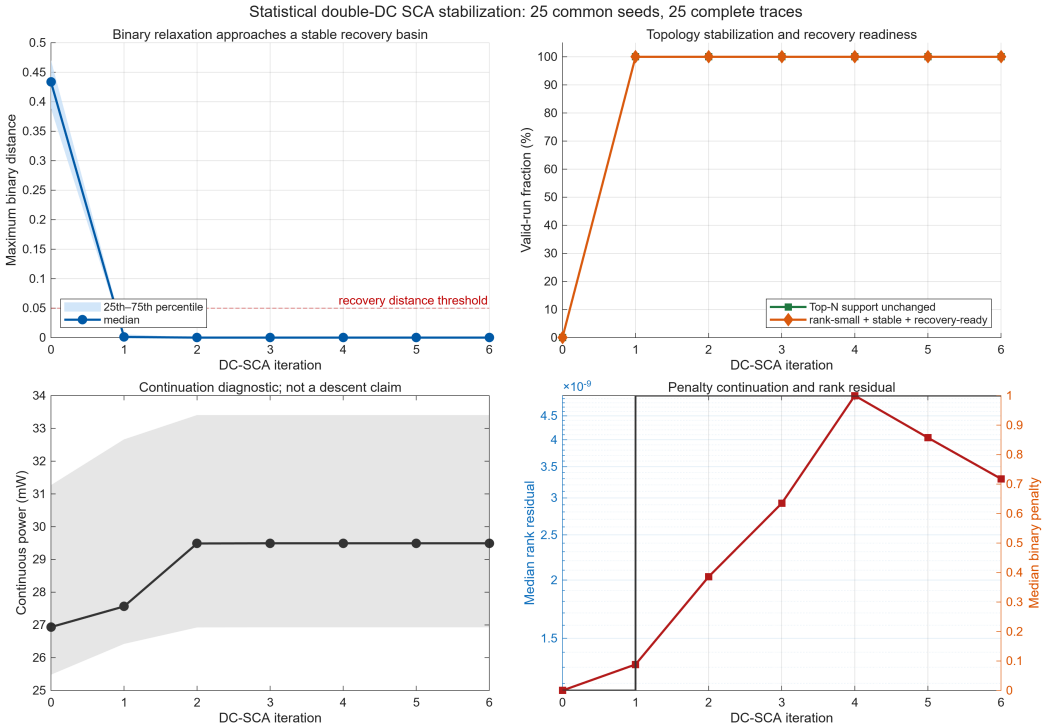


Figure 5.5: Statistical dual-DC SCA stabilization over 25 complete traces. Solid curves report the median across the 25 seeds and shaded bands span the interquartile range (25th–75th percentiles). The recovery-ready statistic combines a small binary distance, a small rank residual, and stable topology support.

5.5 Larger-Scale Validation and Computational Cost

Figure 5.6 examines a larger configuration with $M = 12$, $K = 6$, $P = 3$, and $N_{\text{req}} = 3$. Over eight common seeds, the proposed full recovery is physically feasible in all cases, whereas the FIM-greedy and nearest-AP baselines achieve 75%, and the random topology is feasible in only 12.5% of cases. The

proposed conditional energy cost is close to the FIM-greedy cost and substantially below the nearest-AP and random topologies. Its wall-clock cost is higher than that of fixed-topology baselines because the workflow performs candidate generation, fixed-topology re-solves, and physical validation.

These observations provide scalability evidence at the tested dimensions, not an asymptotic complexity claim. The smaller number of seeds also means that larger independent sweeps and confidence intervals are required before drawing stronger quantitative conclusions.

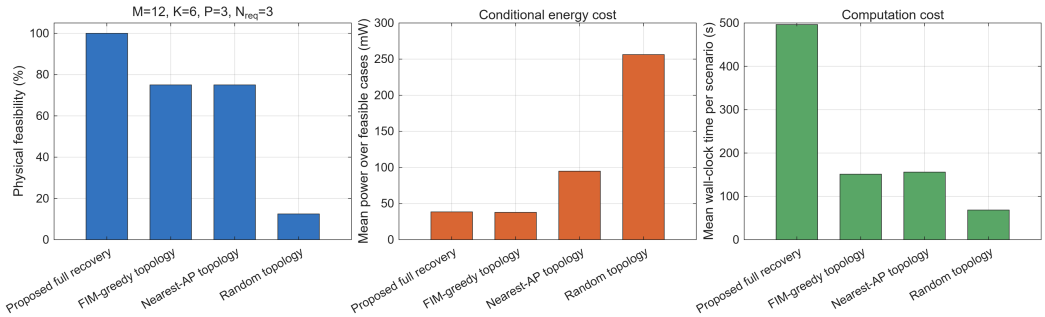


Figure 5.6: Scalability validation for $M = 12$, $K = 6$, $P = 3$, and $N_{\text{req}} = 3$ over eight common seeds. Power and runtime are conditional summaries for physically feasible recovered designs.

5.6 Factor Sweeps, Paired Statistics, and QoS Margins

The comparisons above isolate one mechanism at a time. The extended Monte Carlo campaign stored with the paper result package complements them with a factor-by-factor robustness check: twenty-two configurations organized into six sweep families—AP count $M \in \{4, 6, 8, 10\}$, antennas per AP $N_t \in \{1, 2, 4\}$, UE load $K \in \{2, \ldots, 5\}$, target load $P \in \{1, 2, 3\}$, area side length $\{200, 400, 600\}$ m, and per-AP budget $P_{\max} \in \{17, 20, 23\}$ dBm—together with the two stress geometries of Figure 5.2. Each configuration runs the proposed full recovery, the FIM-greedy topology, and the nearest-AP topology on thirty common seeds, and all statistics follow the feasibility-first protocol of this chapter.

Figure 5.7 summarizes the six families. The proposed recovery is physically feasible in every seed of every non-degenerate configuration, and the FIM-greedy topology matches it. The nearest-AP topology already loses one seed at $M = 4$, $K = 2$, $P = 1$, and in the edge-colocated geometry, loses two

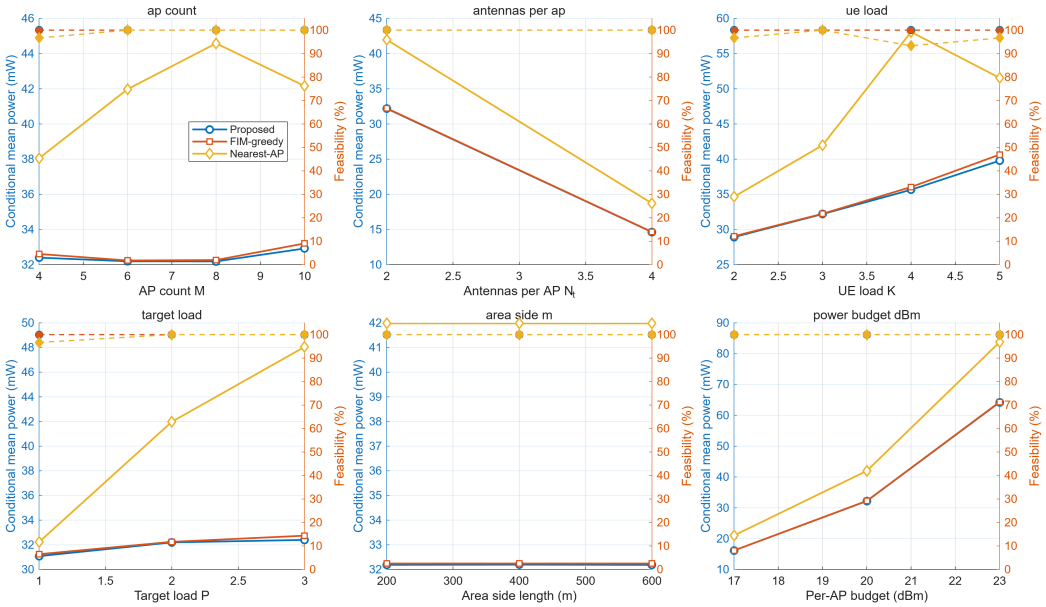


Figure 5.7: Extended factor sweeps over thirty common seeds per configuration. Each panel varies one factor; solid lines (left axis) report conditional mean total power over physically feasible recovered designs, and dashed lines (right axis) report the unconditional physical-feasibility rate. The single-antenna level carries no power marker because no method recovers a feasible design there. In-panel annotations record plotting-pipeline diagnostics and do not modify the constraints defined in Chapter 3.

seeds at $K = 4$, and falls to seven of thirty in the crowded geometry. Conditional mean power of the proposed design stays within 32–33 mW across the whole AP-count range, whereas the nearest-AP topology rises from 38.1 mW at $M = 4$ to 44.6 mW at $M = 8$ and degrades further with load, reaching 58.1 mW at $K = 4$ and 48.1 mW at $P = 3$. Doubling the antennas per AP from two to four halves the conditional power of every method (about 32.2 to 14.6 mW for the proposed design), because per-AP spatial degrees of freedom replace transmit power in meeting the SINR and PCRB constraints. The area side length has essentially no effect over the tested range, indicating that the binding QoS structure is insensitive to geometry spread at fixed node counts.

The single-antenna level is a feasibility boundary rather than a method difference: at $N_t = 1$ all thirty scenarios are rejected at the calibration stage for every method, so no physically feasible design exists. This is consistent with insufficient per-AP spatial degrees of freedom for the simultaneous communication and dedicated-sensing requirements, and it marks $N_t \geq 2$ as a dimension floor for the tested service profile.

One sweep requires care in interpretation. Conditional power scales almost exactly in proportion to the per-AP budget (16.1, 32.2, and 64.2 mW at 17, 20, and 23 dBm). This proportionality is a convention of the scenario generator, which normalizes channel strength so that the best-AP SNR is fixed relative to $P_{\max}$; absolute power levels therefore inherit the budget and must be compared within a configuration, not across budget levels. The sweep nonetheless shows that physical feasibility is preserved across the tested budgets for all three methods.

Table 5.2 quantifies the same-seed differences behind the conditional means. Against the FIM-greedy topology the proposed recovery is within about two per cent in every paired family; several intervals exclude zero, but the gaps stay below 0.9 mW, so the practical difference is negligible at these dimensions. Against the nearest-AP topology the gap is material and grows with load: 18.2% at $M = 8$, 21.3% at $K = 4$, and 19.3% at $P = 3$, rising to 49.6% in the edge-colocated geometry and 81.4% in the crowded geometry. The crowded interval is computed over only the seven seeds where the baseline is also feasible and must be read as a small-sample summary; the paired design nevertheless removes scenario variability, so these intervals are tighter than unpaired comparisons of the same means would suggest.

The QoS audit stored with the campaign explains where the power difference goes. In every feasible configuration the largest mean PCRB ratio stays within 1.0003 of one: the tracking constraint is essentially active at the recovered solutions, so sensing power is provisioned to the required accuracy rather than padded. Consistent with this, the proposed design's worst-link sensing

Table 5.2: Paired same-seed mean power gaps between the proposed recovery and each comparator in representative sweep levels, with 95% confidence intervals obtained from the paired differences. n counts seeds where both compared designs are physically feasible; reduction is the mean per-seed saving relative to the comparator.

Sweep level	Comparator	n	Gap $\pm$ CI half-width (mW)	Reduction (%)
$M = 8$	FIM-greedy	30	0.08 ± 0.08	0.3
$M = 8$	nearest-AP	30	12.39 ± 8.59	18.2
$N_t = 4$	nearest-AP	30	4.08 ± 3.18	13.4
$K = 4$	nearest-AP	28	22.62 ± 17.67	21.3
$K = 5$	FIM-greedy	30	0.86 ± 0.72	1.8
$K = 5$	nearest-AP	29	11.59 ± 6.45	16.6
$P = 3$	FIM-greedy	30	0.34 ± 0.20	1.0
$P = 3$	nearest-AP	30	15.66 ± 11.52	19.3
Edge-colocated	nearest-AP	29	47.76 ± 9.40	49.6
Crowded targets	nearest-AP	7	241.26 ± 19.37	81.4

margin stays within 0.02 dB in all sweeps, while the nearest-AP topology leaves up to 0.08 dB of sensing margin in the ordinary families—power spent on unrequested sensing accuracy. At the same time, the nearest-AP topology keeps a systematically smaller worst-link communication margin (0.26 versus 0.34 dB at $M = 6$, and 0.33 versus 0.44 dB at $P = 3$). The proposed margin grows with AP count, from 0.26 dB at $M = 4$ to 0.74 dB at $M = 10$, and with antennas, reaching 0.96 dB at $N_t = 4$: additional infrastructure converts into QoS headroom instead of additional power.

The failure classification reinforces that infeasibility is scenario-driven rather than recovery-driven. In no configuration does the proposed recovery fail on a seed where another method succeeds. The only configuration in which every method fails jointly is the single-antenna floor; elsewhere, failures concentrate in the nearest-AP topology under crowding and heavy load, exactly where its geometric selection rule is most constrained.

The same audit records the computational price. Median runtime of the proposed workflow grows from 13.1 s at $M = 4$ to 20.2, 37.3, and 67.0 s at $M = 6$, 8, and 10, and reaches 127.2 s at $N_t = 4$—roughly three to five times the fixed-topology baselines at each level. The growth is superlinear in the network dimension, as expected for a workflow that re-solves a convex subproblem for each candidate topology, but the absolute cost remains compatible with offline planning and warm-started re-optimization rather than real-time operation.

5.7 Summary of Findings

The numerical evidence supports four bounded conclusions. First, topology-aware recovery improves physical feasibility relative to nearest and random topology rules in the tested ordinary and pressure geometries. Second, the robust formulation exchanges a modest power increase for substantially improved outage behaviour within the stated CSI-uncertainty model. Third, the dual DC penalties provide recovery-quality evidence, while the larger-scale experiment indicates that the validated workflow remains usable beyond the smallest configuration. Fourth, the factor sweeps show that these findings are stable across AP count, load, area, and budget levels: they expose a dimension floor at one antenna per AP, paired confidence intervals that separate the proposed recovery from the nearest-AP topology but not from the FIM-greedy topology, and a superlinear but planning-compatible runtime growth. These conclusions remain empirical: they do not prove global optimality, universal relaxation tightness, or asymptotic scalability.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis investigated how a cell-free ISAC transmitter can minimize total transmit power while providing robust communication QoS and target-specific sensing service with a physically implementable AP–target topology. The resulting design keeps communication beamforming globally cooperative, while binary association variables authorize only the dedicated sensing covariance of each target. This separation aligns the sensing topology, the target-specific FIM/PCRB requirement, and the dedicated sensing power directly controlled by the optimization.

The formulated mixed-integer problem jointly accounts for worst-case communication SINR, sensing SINR, matrix PCRB, dedicated-sensing authorization, and total AP-power limits. Covariance lifting, the S-Procedure, and a Schur-complement representation yield solver-ready robust and tracking constraints. A dual DC-penalty SCA procedure then reduces both rank-one and binary deficiencies, and direct validation distinguishes a recovered physical design from a relaxed optimization point. The central algorithmic difficulty is that the two discrete structures couple: a rank-one covariance extracted under a wrong association is physically useless, and a binary association read from a high-rank relaxed point need not admit any feasible beamformer. The dual DC formulation exploits the mathematical isomorphism between the rank-deficiency and binarity-deficiency measures and drives both residuals within the same majorization loop, which separates this framework from a plain relaxation-then-rounding pipeline.

The numerical results support four bounded conclusions. First, topology-aware recovery improves physical feasibility relative to nearest-AP and random topology rules in the tested ordinary and pressure geometries. Second,

robust design incurs a modest transmit-power increase while markedly reducing observed outage within the specified CSI-uncertainty model. Third, the dual-DC workflow provides recovery-quality evidence through small rank and binary residuals before physical validation. The larger-dimensional experiment also demonstrates usable recovery behaviour at the tested scale, at a higher computational cost than fixed-topology baselines. Fourth, an extended factor sweep shows that these findings are stable across AP count, antenna, load, area, and budget levels: it exposes a dimension floor at one antenna per AP, and paired same-seed confidence intervals separate the proposed recovery from the nearest-AP topology while staying within about two per cent of the FIM-greedy topology. These are empirical, configuration-specific findings; they do not establish global optimality, universal relaxation tightness, or asymptotic scalability.

These findings also carry practical implications for 6G ISAC deployment. At the hardware level, the dimension floor observed at one antenna per AP shows that joint sensing–communication service cannot be created by software alone: a minimum per-AP spatial resource is a hardware prerequisite that network planning must budget for. At the topology level, the stress-geometry results warn that distance-based sensing clusters—an intuitive rule for distributed networks—fail precisely in the dense layouts where 6G systems are expected to operate, so association decisions must be optimization-driven rather than rule-based. At the computational level, the measured runtime profile positions the workflow as a planning and slow-timescale optimization tool, whose fast-timescale counterpart is identified as future work below.

6.2 Future Work

The immediate priority is broader, reproducible validation. Future experiments should report feasibility, recovered physical transmit power, worst-case communication SINR, sensing SINR, PCRB, per-AP power, and association-cardinality residuals for every sweep, while preserving the distinction between relaxed and recovered solutions. Larger independent seed sets remain needed where the present evidence is thin—the eight-seed larger-scale study and the seven commonly feasible pairs behind the crowded-geometry interval—and systematic PCRB, cluster-size, and CSI-error sweeps should extend the paired confidence intervals reported here, so that the robustness–sensing–power trade-off is quantified more reliably.

Beyond broader validation, three extensions follow naturally from the limitations identified in this thesis. Each is stated together with its motivation and a concrete starting point.

The first extension moves from instantaneous allocation to dynamic target tracking. The formulation of this thesis allocates resources instantaneously within a single scheduling interval, whereas the PCRB requirement is inherently a tracking metric. Practical deployments must follow moving targets such as vehicles and aerial platforms, whose states evolve across intervals, so the present single-snapshot design leaves the temporal structure of the problem unused. A concrete starting point is to augment each target state $\boldsymbol{\theta}_p$ with a kinematic transition model and to propagate the estimation-error covariance by Kalman or extended Kalman filtering, so that $\Gamma_{\mathrm{Track},p}$ arises from a posterior PCRB recursion rather than a fixed single-snapshot bound. When the predicted state is confident, the instantaneous sensing power can be reduced in favour of communication; when prediction confidence drops, the authorized sensing cluster can be enlarged proactively. Casting this allocation sequence as a Markov decision process would finally turn the static design into long-horizon sensing–communication scheduling.

A second extension broadens the robust model from the communication side to the sensing side. The robust design currently protects only the communication side: the norm-bounded uncertainty model applies to the user channels $\hat{\boldsymbol{h}}_k + \Delta\boldsymbol{h}_k$, while the sensing response vector $\boldsymbol{g}_p$ and its derivative matrix $\boldsymbol{D}_p$ are assumed to be exactly known. In practical radar and tracking tasks, however, the target parameters and sensing channels carry estimation errors of their own—radar-cross-section fluctuations and residual clutter are two examples—and these errors propagate into both the sensing-SINR constraint and the FIM/PCRB tracking constraint. A natural approach is to extend the S-Procedure treatment to uncertainty in $(\boldsymbol{g}_p, \boldsymbol{D}_p)$, for example through norm-bounded perturbations of the response and its Jacobian, and to derive robust counterparts of the sensing LMI and the PCRB Schur-complement constraint. The result would be a dual-uncertainty worst-case model in which the robustness guarantee is symmetric between communication and sensing. Environment-aware channel knowledge maps, already exploited for satellite–terrestrial links [41], could provide the prior information needed to calibrate such sensing-side perturbation models.

A third extension concerns the computational scalability of the proposed workflow. The recovery workflow re-solves a convex subproblem for each candidate topology, and the runtime audit of Section 5.6 shows that the computational cost grows superlinearly with the network dimension. This profile is acceptable for planning but not for real-time operation in dense deployments. A promising route is to decompose the central problem across APs with the alternating direction method of multipliers (ADMM) or related federated schemes, so that per-AP power and association constraints are handled locally and only consensus variables travel over the fronthaul.

Candidate pruning or learned surrogates could further reduce the number of fixed-topology re-solves; deep reinforcement learning has already shown promise for beamforming in reconfigurable-antenna systems [42], and the SCA loop can be warm-started from the previous allocation. Fronthaul and processing costs of sensing clusters can be incorporated explicitly at the same time.

Each extension should retain direct validation of the recovered physical variables so that additional modelling flexibility does not obscure the communication, sensing, tracking, and AP-power guarantees.

References

[1] F. Dong, F. Liu, Y. Cui, W. Wang, K. Han, and Z. Wang, "Sensing as a service in 6G perceptive networks: A unified framework for ISAC resource allocation," *IEEE Transactions on Wireless Communications*, vol. 22, no. 5, pp. 3522–3536, May 2023. DOI: 10.1109/TWC.2022.3219463 Accessed: Sep. 18, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9945983/

[2] A. Kaushik et al., "Toward integrated sensing and communications for 6G: Key enabling technologies, standardization, and challenges," *IEEE Communications Standards Magazine*, vol. 8, no. 2, pp. 52–59, Jun. 2024. DOI: 10.1109/MCOMSTD.0007.2300043 Accessed: Oct. 15, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10529727/

[3] F. Liu et al., "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE Journal On Selected Areas in Communications*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022. DOI: 10.1109/JSAC.2022.3156632 Accessed: Sep. 18, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9737357/

[4] F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint radar and communication design: Applications, state-of-the-art, and the road ahead," *IEEE Transactions on Communications*, vol. 68, no. 6, pp. 3834–3862, 2020. DOI: 10.1109/TCOMM.2020.2973976

[5] G. Femenias and F. Riera-Palou, "Scalable cell-free massive MIMO-based integrated sensing and communication," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 10, pp. 15 643–15 659, Oct. 2025. DOI: 10.1109/TVT.2025.3566710 Accessed: Oct. 8, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10982445/

[6] U. Demirhan and A. Alkhateeb, "Cell-free ISAC MIMO systems: Joint sensing and communication beamforming," *IEEE Transactions on Communications*, vol. 73, no. 6, pp. 4454–4468, Jun. 2025. DOI: 10.1109/

TCOMM.2024.3490740 Accessed: Aug. 8, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10742291/

[7] W. Mao, Y. Lu, C.-Y. Chi, B. Ai, Z. Zhong, and Z. Ding, "Communication-sensing region for cell-free massive mimo isac systems," *IEEE Transactions on Wireless Communications*, vol. 23, no. 9, pp. 12 396–12 411, 2024. DOI: 10.1109/TWC.2024.3392330

[8] J. Li et al., "Networked integrated sensing and communications for 6g wireless systems," *IEEE Internet of Things Journal*, vol. 11, no. 17, pp. 29 062–29 075, 2024. DOI: 10.1109/JIOT.2024.3406598

[9] A. Liu et al., "A survey on fundamental limits of integrated sensing and communication," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 994–1034, Sum. 2022. DOI: 10.1109/COMST.2022.3149272 Accessed: Jul. 28, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9705498/

[10] H. Hua, J. Xu, and T. X. Han, "Optimal transmit beamforming for integrated sensing and communication," *IEEE Transactions On Vehicular Technology*, vol. 72, no. 8, pp. 10 588–10 603, Aug. 2023. DOI: 10.1109/TVT.2023.3262513 Accessed: Jul. 28, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10086626/

[11] Z. He, W. Xu, H. Shen, D. W. Kwan Ng, Y. C. Eldar, and X. You, "Integrated sensing and full-duplex communication: Joint transceiver beamforming and power allocation," in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Rhodes Island, Greece: IEEE, Jun. 4, 2023, pp. 1–5. DOI: 10.1109/icassp49357.2023.10097111 Accessed: Jul. 20, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10097111/

[12] K. Cui, H. Chen, J. Zhao, M. Li, and B. Wei, "A review of multi-base station collaborative target sensing based on integrated sensing and communication," *Computer Simulation in Application*, vol. 3, no. 1, pp. 11–17, 2025. DOI: 10.18063/csa.v3i1.913

[13] K. Meng, C. Masouros, A. P. Petropulu, and L. Hanzo, "Cooperative ISAC networks: Opportunities and challenges," *IEEE Wireless Communications*, vol. 32, no. 3, pp. 212–219, Jun. 2025. DOI: 10.1109/MWC.008.2400151 Accessed: Sep. 18, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10726912/

[14] J. Liao, L. Xiang, S. Zhong, L. Xiao, H. Liu, and K. Yang. "Cooperative base station assignment and resource allocation for 6G ISAC network," Accessed: Oct. 2, 2025. [Online]. Available: http://arxiv.org/abs/2509.10240 pre-published.

[15] G. Cheng, Y. Fang, J. Xu, and D. W. K. Ng, "Optimal coordinated transmit beamforming for networked integrated sensing and communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 8, pp. 8200–8214, 2024. DOI: 10.1109/TWC.2023.3346457

[16] Z. Ren, J. Xu, L. Qiu, and D. W. K. Ng, "Secure cell-free integrated sensing and communication in the presence of information and sensing eavesdroppers," *IEEE Journal on Selected Areas in Communications*, vol. 42, no. 11, pp. 3217–3231, 2024. DOI: 10.1109/JSAC.2024.3431582

[17] Z. Yu, X. Hu, C. Liu, M. Peng, and C. Zhong, "Location sensing and beamforming design for IRS-enabled multi-user ISAC systems," *IEEE Transactions on Signal Processing*, vol. 70, pp. 5178–5193, 2022. DOI: 10.1109/TSP.2022.3217353 Accessed: Oct. 20, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9937163/

[18] Y. Y. Chu, M. Shakiba-Herfeh, M. Kamoun, E. Grossi, and S. Buzzi, "Integrated sensing and communication in user-centric cell-free massive MIMO systems with OFDM modulation," in *2023 IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Toronto, ON, Canada: IEEE, Sep. 5, 2023, pp. 1–7. DOI: 10.1109/PIMRC56721.2023.10293932 Accessed: Oct. 21, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10293932/

[19] Z. He, H. Shen, W. Xu, Y. C. Eldar, and X. You, "Mse-based training and transmission optimization for mimo isac systems," *IEEE Transactions on Signal Processing*, vol. 72, pp. 3104–3121, 2024. DOI: 10.1109/TSP.2024.3411641

[20] X. Gong et al., "Digital twin of channel: Diffusion model for sensing-assisted statistical channel state information generation," *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 3805–3821, 2025. DOI: 10.1109/TWC.2025.3542429

[21] S. Hu, Z. Wei, Y. Cai, C. Liu, D. W. K. Ng, and J. Yuan, "Robust and secure sum-rate maximization for multiuser MISO downlink systems with self-sustainable IRS," *IEEE Transactions on Communications*, vol. 69, no. 10, pp. 7032–7049, 2021. DOI: 10.1109/TCOMM.2021.3097140

[22] Y. Geng, T. Hiang Cheng, K. Zhong, K. Chan Teh, and Q. Wu, "Joint beamforming for CRB-constrained IRS-aided ISAC system via product manifold methods," *IEEE Transactions on Wireless Communications*, vol. 24, no. 1, pp. 691–705, Jan. 2025. DOI: 10.1109/TWC.2024.3498105 Accessed: Oct. 20, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10762897/

[23] F. Liu, Y.-F. Liu, A. Li, C. Masouros, and Y. C. Eldar, "Cramér–rao bound optimization for joint radar-communication beamforming," *IEEE Transactions on Signal Processing*, vol. 70, pp. 240–253, 2022. DOI: 10.1109/TSP.2021.3135692

[24] W. Yuan, Z. Wei, S. Li, J. Yuan, and D. W. K. Ng, "Integrated sensing and communication-assisted orthogonal time frequency space transmission for vehicular networks," *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 6, pp. 1515–1528, 2021. DOI: 10.1109/JSTSP.2021.3117404

[25] Q. Lin, H. Shen, Z. Li, W. Xu, C. Zhao, and X. You, "One-bit transceiver optimization for mmwave integrated sensing and communication systems," *IEEE Transactions on Communications*, vol. 73, no. 2, pp. 800–816, 2025. DOI: 10.1109/TCOMM.2024.3440830

[26] X. Wang, J. Li, T. Liu, R. Jiang, and Y. Xu, "Large-scale star-ris assisted near-field isac transmission method," *Journal of Electronics & Information Technology*, vol. 47, no. 1, pp. 147–155, 2025. DOI: 10.11999/JEIT240018

[27] H. Zhang et al., "Holographic integrated sensing and communication," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 7, pp. 2114–2130, 2022. DOI: 10.1109/JSAC.2022.3155548

[28] L. Wang, Y. Cai, X. Li, and S. Hu, "Joint beamforming and port selection for fluid antenna-assisted nearshore ISAC systems," in *Proc. 20th International Symposium on Wireless Communication Systems (ISWCS)*, Accepted for publication, Aug. 2026.

[29] R. Zhao, Y. Zhang, S. Hu, and D. W. K. Ng, "Pinching antenna-assisted ISAC with waveguide mode selection," in *Proc. 20th International Symposium on Wireless Communication Systems (ISWCS)*, Accepted for publication, Aug. 2026.

[30] R. Zhao, S. Hu, D. Mishra, and D. W. K. Ng, "Robust and secure blockage-aware pinching antenna-assisted wireless communication," *IEEE Transactions on Mobile Computing*, 2026, Early Access. DOI: 10.1109/TMC.2026.3695952

[31] S. Hu, R. Zhao, Y. Liao, D. W. K. Ng, and J. Yuan, "Sum-rate maximization for pinching antenna-assisted NOMA systems with multiple dielectric waveguides," in *Proc. IEEE Global Commun. Conf. Workshops (GC Wkshps)*, Dec. 2025, pp. 411–417. DOI: 10.1109/GCWkshps68340.2025.11591116

[32] R. Zhao, S. Hu, D. Mishra, and D. W. K. Ng, "Resource allocation for multi-waveguide pinching-antenna-assisted broadcast networks," in *Proc. IEEE Global Commun. Conf. Workshops (GC Wkshps)*, Dec. 2025, pp. 842–848. DOI: 10.1109/GCWkshps68340.2025.11590996

[33] Z. Zhang, R. Zhao, S. Hu, and D. W. K. Ng, "Energy-efficient resource allocation for six-dimensional movable antenna systems," in *Proc. IEEE 37th Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Accepted for publication, Sep. 2026.

[34] Y. Zhang, S. Hu, D. Mishra, and D. W. K. Ng, "Movable-antenna array-enhanced energy-efficient RSMA communication networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2026.

[35] J. Xu, Y. Cai, Y. Ding, and S. Hu, "Joint resource allocation for relay-assisted nearshore integrated sensing and communication systems," in *Proc. 20th International Symposium on Wireless Communication Systems (ISWCS)*, Accepted for publication, Aug. 2026.

[36] X. Li, Z. Wu, Y. Cai, S. Hu, Y. Liao, and W. Yuan, "Win-win of communication and sensing security for MC-NOMA ISAC systems," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 9, pp. 3214–3230, 2025. DOI: 10.1109/JSAC.2025.3574620

[37] F. Peng, Z. Jiang, S. Zhou, Z. Niu, and S. Zhang, "Sensing and communication co-design for status update in multiaccess wireless networks," *IEEE Transactions on Mobile Computing*, vol. 22, no. 3, pp. 1779–1792, 2023. DOI: 10.1109/TMC.2021.3108544

[38] Y. Cai, Z. Wei, S. Hu, C. Liu, D. W. K. Ng, and J. Yuan, "Resource allocation and 3D trajectory design for power-efficient IRS-assisted UAV-NOMA communications," *IEEE Transactions on Wireless Communications*, vol. 21, no. 12, pp. 10 315–10 334, 2022. DOI: 10.1109/TWC.2022.3183300

[39] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, United Kingdom: Cambridge University Press, 2004.

[40] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye, and S. Zhang, "Semidefinite relaxation of quadratic optimization problems," *IEEE Signal Processing Magazine*, vol. 27, no. 3, pp. 20–34, 2010. DOI: 10.1109/MSP.2010.936019

[41] T. Shen, X. Wu, Y. Zhan, S. Hu, D. W. K. Ng, and L. Yang, "Environment-aware satellite-terrestrial channel knowledge map," in *Proc. 20th Int. Symp. Wireless Commun. Syst. (ISWCS)*, Accepted for publication, Aug. 2026.

[42] M. Du, S. Hu, C. Liu, and D. W. K. Ng, "Beamforming for RIS-assisted pinching antenna systems: A deep reinforcement learning approach," in *Proc. IEEE 27th Int. Workshop Signal Process. Artif. Intell. Wireless Commun. (SPAWC)*, Accepted for publication, Sep. 2026.